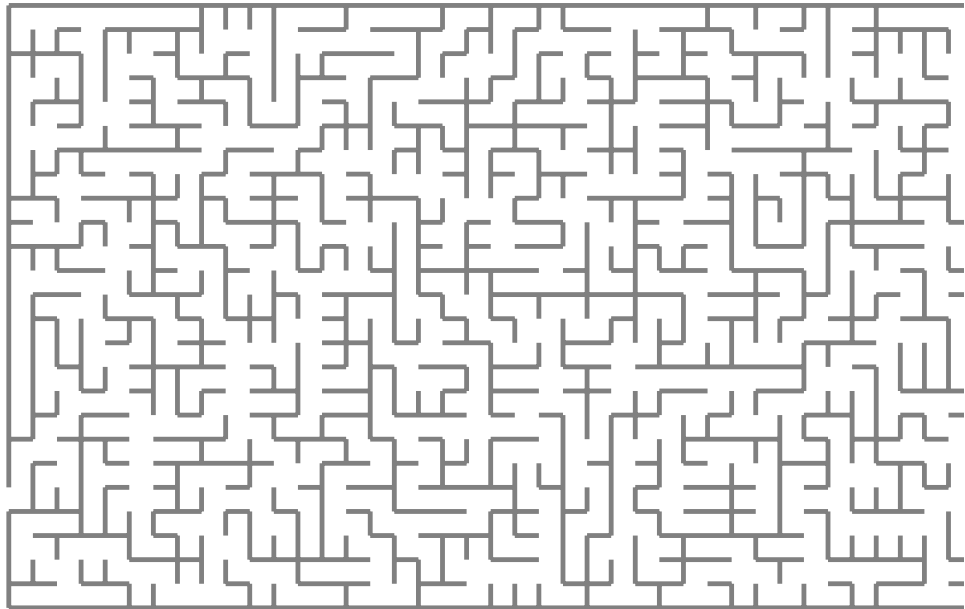


Training Dynamics of
Overparameterized Neural Networks:
A Study of Spectral Bias on S^1

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Shreyas Kalvankar

Training Dynamics of Overparameterized Neural Networks: A Study of Spectral Bias on S^1

THESIS

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by

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Training Dynamics of Overparameterized Neural Networks: A Study of Spectral Bias on S^1

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Abstract

This thesis studies the training dynamics of overparameterized neural networks through spectral bias, the tendency of gradient-based training to learn low-frequency components of a target function faster than high-frequency ones. The setting is regression on the unit circle using a one-hidden-layer ReLU network, with targets given by finite Fourier mixtures.

The analysis begins in the infinite-width regime. In this limit, the operator governing the residual dynamics is the neural tangent kernel. Under the uniform measure on the circle, this operator is a convolution and is therefore diagonalized by the Fourier basis. This gives an explicit description of the residual dynamics and of the corresponding frequency-dependent learning rates.

The main contribution of the current thesis is a finite-sample analysis of this continuum operator. On a fixed low-frequency Fourier subspace, non-asymptotic bounds are derived that show the sampled operator remains close to the continuum operator with high probability. The final part of the project, still in progress, studies the effect of finite width in the continuum setting.

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Preface

This is where you thank people for helping you etc.

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May 15, 2026

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Chapter 1

Introduction

Deep neural networks have achieved remarkable empirical success across a wide range of application domains, most notably in computer vision and natural language processing. In vision, deep convolutional and residual architectures have led to dramatic improvements on large-scale benchmarks, establishing deep learning as the dominant paradigm for visual recognition tasks (Krizhevsky et al., 2012; He et al., 2016). Similarly, in natural language processing, attention-based architectures introduced by Vaswani et al. (2023) such as Transformers have enabled substantial advances in sequence modeling and representation learning, and now form the foundation of modern large-scale language models. Over the past decade, this empirical progress has been accompanied by a steady increase in model size, depth, and computational scale, often resulting in highly overparameterized models that generalize well despite their capacity to fit random labels (Zhang et al., 2017; Belkin et al., 2019). Much of this progress has been driven by empirical experimentation and architectural intuition rather than by a complete theoretical understanding of the mechanisms underlying training and generalization in deep neural networks.

Despite their practical success, the mechanisms governing the training dynamics and convergence behavior of neural networks remain only partially understood, even in highly simplified settings. From an optimization perspective, neural network training involves high-dimensional, non-convex objectives whose geometry depends intricately on architectural choices, initialization, and optimization dynamics. Early analyses of neural network loss landscapes have highlighted the prevalence of saddle points and complex critical structures, underscoring the difficulty of directly characterizing training dynamics in parameter space Choromanska et al. (2015). While more recent theoretical work has made progress in analyzing overparameterized models under restrictive assumptions such as two-layer networks or specific scaling regimes these results do not yet provide a unified explanation of neural network training in general settings Arora et al. (2019). More broadly, this gap between empirical performance and theoretical understanding has motivated the study of neural networks through simplified models and asymptotic limits, where training dynamics can be analyzed more precisely.

Structure of the thesis. This thesis aims to contribute to the theoretical study of neural-network training by focusing on one such simplified setting in which the learning dynamics

can be analyzed explicitly. We begin by introducing the supervised-learning framework and empirical risk minimization, which provide the basic language for describing how models are trained from data. We then consider increasingly expressive model classes, moving from linear models to deep linear networks and nonlinear neural networks, to show where training dynamics can be characterized cleanly and where this understanding begins to break down. This progression motivates the use of simplified theoretical regimes: settings that are restrictive enough to permit precise analysis, but still rich enough to capture phenomena observed in neural-network training. After this foundation, we review existing theoretical approaches for studying wide neural networks. This discussion naturally leads to spectral bias, the empirical and theoretical observation that neural networks often learn smooth, low-frequency components of a target function before more oscillatory, high-frequency components. Spectral bias is the main object of study in this thesis. The related work then summarizes existing empirical and theoretical explanations of this phenomenon, leading to the research gap and the research questions addressed in the remainder of the thesis.

1.1 Problem setup

The Statistical learning framework The fundamental objective in supervised learning is to identify a functional relationship between an input space $\mathcal{X} \subseteq \mathbb{R}^d$ and a target space $\mathcal{Y}$. We assume the existence of a joint probability distribution $\mathbf{p}$ over $\mathcal{X} \times \mathcal{Y}$. Ideally, one seeks a measurable function $f : \mathcal{X} \rightarrow \mathcal{Y}$, that minimizes the expected risk (or population risk):

$$\mathcal{L}_{\mathbf{p}}(f) = \mathbb{E}_{\mathbf{p}} \ell(f(x), y) \quad (1.1)$$

where $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, \infty]$ is a loss function quantifying the discrepancy between the prediction and the true target. In practice, two fundamental constraints prevent the direct minimization of the equation (1.1). First, searching over the space of all possible measurable functions is computationally and analytically intractable, so we usually restrict our search to a specific hypothesis class which is a family of functions f_{θ} parameterized by a vector $\theta \in \mathbb{R}^p$. Second, the underlying distribution $\mathbf{p}$ is unknown; we only have access to a finite training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ of n samples drawn i.i.d from $\mathbf{p}$.

Empirical risk minimization. The second constraint leads us to the principle of empirical risk minimization (ERM). Given a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, a measurable function f_{θ} parameterized by θ , and a loss function $\ell(\cdot, \cdot)$, ERM seeks parameters that minimize the empirical risk,

$$\theta^* \in \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i). \quad (1.2)$$

This formulation isolates the key ingredients that determine the behavior of learning algorithms: the function class induced by the parameterization $\theta \mapsto f_{\theta}$, the geometry of the loss landscape, and the optimization procedure used to minimize the empirical risk. In modern neural network training, the empirical risk is typically minimized using gradient-based methods, which induce a dynamical system in parameter space whose properties depend intricately on both the model architecture and the chosen loss function.

Linear models and gradient descent. To build intuition, we first consider the case of linear models. We define a function $f_w(x) : \mathbb{R}^d \rightarrow \mathbb{R}$, parameterized by $w \in \mathbb{R}^d$ with the prescription $f_w(x) = x^\top w$. We specialize the loss function to the squared error, $\ell(f(x), y) = \frac{1}{2}(f_w(x) - y)^2$. Let $X \in \mathbb{R}^{d \times n}$ denote the data matrix whose columns are x_i , and let $y \in \mathbb{R}^n$ denote the vector of targets. With slight abuse of notation, the ERM problem simplifies to a least-squares objective:

$$\arg \min_w \mathcal{L}(w) = \frac{1}{2} \sum_{i=1}^n (f_w(x_i) - y_i)^2 = \frac{1}{2} \|X^\top w - y\|_2^2. \quad (1.3)$$

This objective is convex and differentiable, with gradient

$$\nabla_w \mathcal{L}(w) = X(X^\top w - y) = XX^\top w - Xy. \quad (1.4)$$

At a stationary point w^* , the gradient vanishes, yielding the equations

$$XX^\top w^* = Xy, \quad (1.5)$$

which characterize the set of global minimizers of $\mathcal{L}$.

The structure of the minimizers depends on the rank of the empirical covariance operator $A := XX^\top$. If A is positive definite (equivalently, if X has full row rank), then the minimizer is unique and given by $w^* = A^{-1}Xy$. In contrast, in the overparameterized regime where $\text{rank}(X) < d$, the matrix A is singular and the least-squares objective admits infinitely many global minimizers. In this case, the Moore–Penrose pseudoinverse can be used to approximate a solution. Since A is symmetric, its pseudoinverse A^+ acts as the inverse on $\text{range}(A)$ and annihilates $\ker(A)$, yielding the minimum-norm least-squares solution $w^* = A^+Xy$.

Gradient descent with step size $\eta > 0$ takes the explicit form

$$w_{t+1} = w_t - \eta \nabla_w \mathcal{L}(w_t) = (I - \eta XX^\top) w_t + \eta Xy. \quad (1.6)$$

Assuming $w_0 = 0$, we can write a closed form solution

$$w_t = \eta \sum_{s=0}^{t-1} (I - \eta XX^\top)^s Xy. \quad (1.7)$$

Let $\lambda_{\max}(A)$ denote the largest eigenvalue of A . If the step size satisfies $\eta \in (0, 2/\lambda_{\max}(A))$, then for any least-squares minimizer w^* the error $e_t := w_t - w^*$ evolves as

$$e_{t+1} = (I - \eta A)e_t. \quad (1.8)$$

Since A is symmetric, it admits an eigendecomposition

$$A = Q\Lambda Q^\top,$$

where $Q \in \mathbb{R}^{d \times d}$ is orthogonal ($Q^\top Q = I$) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ contains the (real) eigenvalues of A . Moreover, A is positive semidefinite because $v^\top A v = \|X^\top v\|_2^2 \geq 0$ for all v , hence $\lambda_i \geq 0$ and in particular $\lambda_i \in [0, \lambda_{\max}(A)]$.

1. INTRODUCTION

Starting from the error recursion

$$e_{t+1} = (I - \eta A)e_t,$$

we express the error in the eigenbasis of A by defining $\tilde{e}_t := Q^\top e_t$. Left-multiplying by $Q^\top$ and using $e_t = Q\tilde{e}_t$ gives

$$\tilde{e}_{t+1} = Q^\top (I - \eta A) Q \tilde{e}_t = (I - \eta Q^\top A Q) \tilde{e}_t = (I - \eta \Lambda) \tilde{e}_t.$$

Since Λ is diagonal, this yields the component-wise dynamics

$$\tilde{e}_{t+1,i} = (1 - \eta \lambda_i) \tilde{e}_{t,i}, \quad i = 1, \dots, d. \quad (1.9)$$

If $\eta \in (0, 2/\lambda_{\max}(A))$, then for any $\lambda_i > 0$ we have $0 < \eta \lambda_i \leq \eta \lambda_{\max}(A) < 2$, hence $-1 < 1 - \eta \lambda_i < 1$, i.e. $|1 - \eta \lambda_i| < 1$. Under the stated condition on η , we have $|1 - \eta \lambda_i| < 1$ for all $\lambda_i > 0$, implying geometric decay of all components of e_t in $\text{range}(A)$. In particular, if A is positive definite, then $w_t \rightarrow w^*$ as $t \rightarrow \infty$. More generally, if $w_0 \in \text{range}(A)$ (for example, $w_0 = 0$), then w_t converges to the minimum-norm least-squares solution $w^* = A^+ X y$.

The eigen-decomposition above shows that gradient descent acts as a linear dynamical system whose behavior is governed by the spectrum of the empirical covariance operator $XX^\top$. Directions corresponding to larger eigenvalues converge more rapidly, while components in the null space of $XX^\top$ remain unchanged. In the overparameterized regime, where multiple *interpolating solutions* exist, i.e., solutions satisfying $X^\top w = y$, gradient descent from standard initializations converges to a particular interpolating solution, namely the minimum-norm solution. This illustrates how the optimization algorithm induces an implicit bias even in this simplest linear setting.

This simple example already contains the basic spectral mechanism that will reappear throughout the thesis: gradient descent does not reduce all components of the error at the same speed; instead, the decay rate of each component is determined by the eigenvalue of the operator governing the dynamics. In the linear model above, these eigendirections are determined by the empirical covariance matrix $XX^\top$, so they do not yet have to correspond to frequencies.

Deep linear network. Even in the absence of nonlinear activation functions, introducing an additional layer already leads to substantially more intricate training dynamics. Consider a two-layer linear network with scalar output,

$$f_{v,W}(x) := v^\top W x, \quad (1.10)$$

where $W \in \mathbb{R}^{m \times d}$, $v \in \mathbb{R}^m$, and m denotes the width of the hidden layer. The least-squares empirical risk is

$$\mathcal{L}(v, W) := \frac{1}{2} \|X^\top W^\top v - y\|_2^2. \quad (1.11)$$

Although the resulting function is linear in the input, the loss is no longer convex in the parameters (v, W) due to their multiplicative coupling.

Let

$$r := X^\top W^\top v - y \in \mathbb{R}^n \quad (1.12)$$

denote the residual vector. The gradients of $\mathcal{L}$ are given by

$$\nabla_v \mathcal{L} = W X r, \quad \nabla_W \mathcal{L} = v(Xr)^\top. \quad (1.13)$$

Similar to the linear case, the parameters of this model are updated iteratively using gradient descent with step size $\eta > 0$

$$v_{k+1} = v_k - \eta \nabla_v \mathcal{L}(v_k, W_k), \quad W_{k+1} = W_k - \eta \nabla_W \mathcal{L}(v_k, W_k).$$

The analysis of learning dynamics is often simplified by considering the infinitesimal learning rate limit ($\eta \rightarrow 0$). In this regime, the discrete updates converge to a system of coupled ordinary differential equations (ODEs) known as gradient flow.

$$\dot{v}_t = -\nabla_v \mathcal{L}(v_t, W_t) = W_t X (y - X^\top W_t^\top v_t), \quad \dot{W}_t = -\nabla_W \mathcal{L}(v_t, W_t) = v_t (y - X^\top W_t^\top v_t)^\top X^\top. \quad (1.14)$$

This coupled system consists of a vector-valued and a matrix-valued ordinary differential equation and is nonlinear in the parameters, despite the underlying predictor being linear in the input. As a result, the training dynamics no longer admit a simple spectral characterization as in the one-layer case.

Nevertheless, recent work has shown that deep linear networks admit a more structured description in suitable asymptotic regimes. In particular, Chizat et al. (2024) study gradient flow for deep linear networks in an infinite-width limit, where the dynamics are described at the level of a measure-valued evolution. While no closed-form solution of the finite-dimensional ODE system is obtained, this framework establishes global well-posedness, convergence to global minimizers, and the emergence of implicit regularization effects induced by depth.

Nonlinear networks. The analysis of training dynamics becomes more involved once nonlinear activation functions are introduced. Consider again a two-layer network of the form

$$f_{v,W}(x) = v^\top \phi(Wx),$$

where $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is a continuously differentiable activation function applied elementwise. For the least-squares objective, the continuous-time evolution of the parameters is governed by the gradient flow system:

$$\dot{v}_t = \phi(W_t X) (y - \phi(W_t X)^\top v_t), \quad \dot{W}_t = \left(\phi(W_t X) \odot \left(v_t (y - \phi(W_t X)^\top v_t)^\top \right) \right) X^\top, \quad (1.15)$$

where $\odot$ denotes the Hadamard (elementwise) product. The gradient flow equations involve both the activations $\phi(Wx)$ and their derivatives $\phi'(Wx)$. In particular, the evolution of the parameters mixes standard matrix products with elementwise nonlinear operations. As a consequence, the training dynamics of nonlinear networks do not admit a simple closed-form or spectral description in general.

Nevertheless, the linear example suggests the kind of structure one would like to recover: an operator-level description in which different components of the error decay at different rates. For nonlinear neural networks, identifying such an operator and understanding its spectrum is substantially more difficult, because the relevant dynamics depend on the evolving parameters of the network. This is one reason why studying frequency-dependent learning, or spectral bias, is non-trivial in general neural networks. The remainder of the thesis therefore focuses on simplified regimes in which an operator governing the dynamics can be made explicit, allowing the connection between its spectrum and the observed learning behavior to be analyzed more precisely.

1.2 Thesis motivation.

At a high level, this thesis is motivated by the question of how these networks' training dynamics can be studied in a mathematically tractable setting without discarding the phenomena one ultimately seeks to understand. The objective is not to provide a complete description of neural network training in full generality, but rather to identify a setting in which the dynamics can be analyzed explicitly while still retaining features that remain meaningful for finite networks and finite datasets.

This motivates the study of analytical regimes in which the dynamics become tractable. Although such regimes necessarily involve simplifications like taking large-width limits, they can still provide useful insight into mechanisms that are otherwise difficult to access in the full non-linear parameter dynamics. Two prominent examples of such approaches are the neural tangent kernel and mean-field formulations, which offer complementary perspectives on the behavior of wide neural networks during training.

In this thesis, that regime is a single-hidden-layer ReLU network with inputs on the unit circle S^1 . In the infinite-width limit, the NTK defines a deterministic integral operator. For the uniform distribution on the circle, this operator is diagonalized by the Fourier basis. This gives a clean reference picture for spectral bias: each Fourier mode has its own learning rate. The rest of the thesis studies how this picture changes at finite sample size and finite width.

Chapter 2

Related Work

2.1 Neural Tangent Kernel

As discussed in Chapter 1, for nonlinear neural networks the gradient flow dynamics are parameter-dependent and do not admit a simple closed-form description. One approach to recovering a tractable analytical framework is to consider regimes in which the network behaves approximately linearly around its random initialization. The *Neural Tangent Kernel* (NTK), introduced by Jacot et al. (2020), formalizes this idea by studying training dynamics through a first-order linearization in parameter space, leading to a kernel-based description of learning in wide neural networks.

Linearization around initialization Let $f(x; \theta)$ denote a neural network with parameters $\theta \in \mathbb{R}^p$, initialized at θ_0 . A first-order Taylor expansion around θ_0 yields

$$f(x; \theta) \approx f(x; \theta_0) + \nabla_{\theta} f(x; \theta_0)^{\top} (\theta - \theta_0), \quad (2.1)$$

where $f(x; \theta_0)$ is the network output at initialization and

$$\phi(x) := \nabla_{\theta} f(x; \theta_0)$$

defines the tangent feature map. Locally around initialization, the network thus behaves as a linear model in parameter space,

$$f(x; \theta) \approx f(x; \theta_0) + \phi(x)^{\top} (\theta - \theta_0). \quad (2.2)$$

Definition 2.1 (Neural Tangent Kernel (Jacot et al., 2020)). Given an initialization θ_0 , the neural tangent kernel is defined as

$$\Theta_0(x, x') := \nabla_{\theta} f(x; \theta_0)^{\top} \nabla_{\theta} f(x'; \theta_0).$$

The NTK measures how infinitesimal parameter updates couple the network outputs at different inputs and plays the role of an empirical covariance operator in the tangent feature space.

2. RELATED WORK

Training dynamics induced by the NTK We consider training with the squared loss using gradient descent with step size $\eta > 0$,

$$L(\theta) = \frac{1}{2} \sum_{i=1}^n (f(x_i; \theta) - y_i)^2, \quad \theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t), \quad t = 0, 1, 2, \dots$$

Define $f_t(x) := f(x; \theta_t)$. The gradient of the loss is

$$\nabla_{\theta} L(\theta_t) = \sum_{j=1}^n (f_t(x_j) - y_j) \nabla_{\theta} f(x_j; \theta_t).$$

Applying a first-order Taylor expansion of $f(\cdot; \theta)$ around θ_t gives

$$f_{t+1}(x_i) \approx f_t(x_i) + \nabla_{\theta} f(x_i; \theta_t)^{\top} (\theta_{t+1} - \theta_t).$$

Substituting the gradient descent update $\theta_{t+1} - \theta_t = -\eta \nabla_{\theta} L(\theta_t)$ yields

$$f_{t+1}(x_i) = f_t(x_i) - \eta \sum_{j=1}^n \nabla_{\theta} f(x_i; \theta_t)^{\top} \nabla_{\theta} f(x_j; \theta_t) (f_t(x_j) - y_j),$$

which can be written compactly as

$$f_{t+1}(x_i) = f_t(x_i) - \eta \sum_{j=1}^n \Theta_t(x_i, x_j) (f_t(x_j) - y_j), \quad (2.3)$$

where the iteration-dependent NTK is

$$\Theta_t(x_i, x_j) := \nabla_{\theta} f(x_i; \theta_t)^{\top} \nabla_{\theta} f(x_j; \theta_t).$$

If we stack predictions into $f_t := (f_t(x_1), \dots, f_t(x_n))^T \in \mathbb{R}^n$, in vector form, the dynamics read

$$f_{t+1} = f_t - \eta \Theta_t (f_t - y). \quad (2.4)$$

where $\bar{\Theta} = (\bar{\Theta}(x_i, x_j))_{i,j=1}^n \in \mathbb{R}^{n \times n}$ is the kernel gram matrix. In general, Θ_t evolves during training, reflecting changes in the network's tangent features.

One-hidden-layer NTK and infinite-width limit To make the NTK explicit, consider a two-layer network of width m ,

$$f(x) = \frac{1}{\sqrt{m}} \sum_{\alpha=1}^m v_{\alpha} \varphi(w_{\alpha}^{\top} x), \quad v_{\alpha} \in \mathbb{R}, \quad w_{\alpha}, x \in \mathbb{R}^d. \quad (2.5)$$

Each neuron is parameterized by $\theta_{\alpha} = (v_{\alpha}, w_{\alpha})$ and randomly initialized as

$$v_{\alpha} \sim \mathcal{N}(0, \sigma_v^2), \quad w_{\alpha} \sim \mathcal{N}(0, \sigma_w^2 I_d),$$

independently across $\alpha = 1, \dots, m$. $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a non-linear activation function.

Finite-width NTK at initialization. By direct computation,

$$\nabla_{v_\alpha} f(x) = \frac{1}{\sqrt{m}} \varphi(w_\alpha^\top x), \quad (2.6)$$

$$\nabla_{w_\alpha} f(x) = \frac{1}{\sqrt{m}} v_\alpha \varphi'(w_\alpha^\top x) x. \quad (2.7)$$

Substituting into the definition of the NTK yields the empirical kernel

$$\Theta_0^{(m)}(x, x') = \frac{1}{m} \sum_{\alpha=1}^m \varphi(w_\alpha^\top x) \varphi(w_\alpha^\top x') + \frac{1}{m} \sum_{\alpha=1}^m v_\alpha^2 \varphi'(w_\alpha^\top x) \varphi'(w_\alpha^\top x') x^\top x'. \quad (2.8)$$

Infinite-width limit. Each sum in (2.8) is an empirical average of i.i.d. terms. By the strong law of large numbers,

$$\Theta_0^{(m)}(x, x') \xrightarrow{\text{a.s.}} \bar{\Theta}(x, x') \quad \text{as } m \rightarrow \infty,$$

where the limiting NTK is deterministic and given by

$$\bar{\Theta}(x, x') = \mathbb{E}_w[\varphi(w^\top x) \varphi(w^\top x')] + \sigma_v^2 x^\top x' \mathbb{E}_w[\varphi'(w^\top x) \varphi'(w^\top x')]. \quad (2.9)$$

This argument extends to deep fully connected networks, where both the forward covariance kernel and the NTK satisfy recursive layerwise equations in the infinite-width limit (Jacot et al., 2020; Lee et al., 2020).

2.1.1 Constant-kernel regime and spectral dynamics

In the infinite-width limit, the NTK remains constant throughout training, $\Theta_t \equiv \bar{\Theta}$. Equation (2.4) then reduces to the linear iteration

$$f_{t+1} = f_t - \eta \bar{\Theta}(f_t - y). \quad (2.10)$$

Letting $r_t := f_t - y$, we obtain

$$r_{t+1} = (I - \eta \bar{\Theta}) r_t.$$

Assuming $\eta < 2/\lambda_{\max}(\bar{\Theta})$, the residual admits the closed form

$$r_t = (I - \eta \bar{\Theta})^t r_0, \quad f_t = y + (I - \eta \bar{\Theta})^t (f_0 - y). \quad (2.11)$$

As in the linear setting of Chapter 1, convergence occurs independently along the eigenvectors of $\bar{\Theta}$, with geometric rates determined by the corresponding eigenvalues.

NTK dynamics, solution structure, and implicit bias In the constant-kernel (infinite-width) regime, the NTK prediction dynamics reduce to a linear iteration in prediction space. Eq 2.10, admits fixed points f^* satisfying

$$\bar{\Theta}(f^* - y) = 0. \quad (2.12)$$

2. RELATED WORK

Equation (2.12) characterizes the set of global minimizers of the squared loss in the NTK regime. As in linear least squares, the structure of this solution set depends on the rank of the operator $\bar{\Theta}$.

If $\bar{\Theta}$ is strictly positive definite on the training set, the solution is unique and satisfies $f^* = y$. When $\bar{\Theta}$ is singular, the loss admits infinitely many interpolating solutions in prediction space, differing by elements of $\ker(\bar{\Theta})$. In this case, gradient descent converges to a particular fixed point determined by the initialization.

Tangent features and operator factorization. To formulate the NTK in a way that remains meaningful in the infinite-width limit, it is convenient to view parameter perturbations as elements of a Hilbert space $\mathcal{H}_\theta$ equipped with an inner product $\langle \cdot, \cdot \rangle$. In the finite-dimensional case, $\mathcal{H}_\theta = \mathbb{R}^p$ with the Euclidean inner product; in the infinite-width limit, $\mathcal{H}_\theta$ denotes the corresponding limit space of parameter perturbations, often referred to as the tangent parameter space (Jacot et al., 2020; Lee et al., 2020).

In this setting, the limiting NTK Gram matrix on the training set can be expressed as

$$\bar{\Theta} = J_0 J_0^*, \quad (2.13)$$

where $J_0 : \mathcal{H}_\theta \rightarrow \mathbb{R}^n$ denotes the Jacobian of the network outputs with respect to parameters at initialization, viewed as a linear operator,

$$(J_0 h)_i = \langle \nabla_{\theta} f(x_i; \theta_0), h \rangle,$$

and J_0^* is its adjoint. This representation makes explicit that $\bar{\Theta}$ is symmetric and positive semidefinite, and that it acts as an empirical covariance operator for the tangent features induced by the network at initialization (Jacot et al., 2020).

Implicit bias and representer viewpoint. When $\bar{\Theta}$ is singular, the Moore–Penrose pseudoinverse $\bar{\Theta}^+$ acts as the inverse on $\text{range}(\bar{\Theta})$ and annihilates $\ker(\bar{\Theta})$. Consequently, constant-kernel NTK training eliminates only the residual component lying in $\text{range}(\bar{\Theta})$ while preserving the component in $\ker(\bar{\Theta})$. This behavior reflects an implicit regularization effect analogous to linear least squares and can be interpreted as convergence to a minimum-norm solution in the reproducing kernel space associated with $\bar{\Theta}$, as formalized by representer-theorem results (Bietti and Mairal, 2019; Bartolucci et al., 2021).

Limitations of stable kernels and feature learning. Although the NTK framework provides a clean and tractable description of training dynamics, the stability of the kernel also introduces inherent limitations. When the kernel does not change during training, learning is effectively confined to a fixed feature space. Indeed, any positive semidefinite kernel admits a representation $K(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{H}}$, so that optimization in the NTK regime corresponds to fitting a linear model in the associated reproducing kernel Hilbert space $\mathcal{H}$.

In contrast, for finite-width networks the NTK typically evolves during training, implying that the tangent features $\Phi_t(x) = \nabla_{\theta} f(x; \theta_t)$ also change over time. This evolution allows the network to adapt its representation to the data, a phenomenon commonly referred

to as *feature learning*. Such effects are absent in a strictly stable-kernel regime, where only the coefficients of fixed features are adjusted.

The importance of feature learning is already visible in simplified settings. Even purely linear networks exhibit nontrivial training dynamics due to the coupling of parameters across layers, leading to implicit biases that cannot be captured by a fixed kernel model (Saxe et al., 2014; Tu et al., 2024). From this viewpoint, the NTK regime can be seen as a useful but restrictive approximation, which motivates studying regimes where kernel evolution and feature learning play an explicit role.

2.2 Mean-field representation of a two-layer network

While the neural tangent kernel provides a linearized description of training dynamics around initialization, it does not capture regimes in which features evolve significantly during optimization. An alternative and complementary approach is provided by the *mean-field* perspective, which studies wide neural networks by viewing their parameters as interacting particles and describing training as the evolution of a probability distribution over parameter space.

As the network width tends to infinity under appropriate scalings, the training dynamics induced by gradient descent converge to a deterministic evolution equation for the parameter distribution. This viewpoint was first formalized for two-layer networks by Mei et al. (2018), who derived a nonlinear partial differential equation governing the evolution of the parameter measure. Related interacting particle system formulations and convergence results were developed by Rotskoff and Vanden-Eijnden (2022), while extensions to multi-layer networks were proposed by Nguyen (2019). From an optimization perspective, Chizat and Bach (2018) showed that, under suitable initialization and in the many-particle limit, training dynamics can be interpreted as a Wasserstein gradient flow on the space of probability measures. Informally, this means that instead of tracking individual neurons, one tracks how their overall distribution moves over parameter space during training. The evolution is governed by a transport equation, so learning is viewed as a continuous redistribution of mass that decreases the loss. In this formulation, the evolution of the network is described directly at the level of measures, and the flow converges to global minimizers despite the non-convexity of the finite-dimensional parameterization.

A detailed measure-theoretic derivation of the mean-field representation is out of the scope of this thesis and is deferred to Appendix A.1.4. Here we only record the aspects that are relevant for contrasting the mean-field viewpoint with the NTK regime.

Consider the two-layer neural network from (2.5) with modified scaling of $\frac{1}{m}$

$$f_{\theta}(x) = \frac{1}{m} \sum_{\alpha=1}^m v_{\alpha} \varphi(w_{\alpha}^{\top} x), \quad v_{\alpha} \in \mathbb{R}, w_{\alpha}, x \in \mathbb{R}^d, \quad (2.14)$$

where each neuron is parameterized by $\theta_{\alpha} = (v_{\alpha}, w_{\alpha})$, $\alpha \in \{1, \dots, m\}$.

2. RELATED WORK

Integral representation and empirical measure Let $\Omega := \mathbb{R} \times \mathbb{R}^d$ denote the parameter space, and define the empirical neuronal measure

$$\mu_m := \frac{1}{m} \sum_{\alpha=1}^m \delta_{(v_\alpha, w_\alpha)}. \quad (2.15)$$

Then $\mu_m \in \mathcal{P}(\Omega)$, where $\mathcal{P}(\Omega)$ is the space of Borel probability measures on Ω . A detailed derivation is given in Appendix A.1.1.

With this notation, the network output can be written as

$$f_\theta(x) = \int_{\Omega} v \Phi(w^\top x) d\mu_m(v, w). \quad (2.16)$$

Thus, under the mean-field parameterization, the network depends on its parameters only through their empirical distribution.

Mean-field viewpoint. Equation (2.16) shows that the network output can be written as a linear functional of the empirical measure μ_m . This representation is only possible because we scale the output by $1/m$ and because the hidden neurons are permutation invariant. Such observations motivate the mean-field viewpoint, in which the empirical measure μ_m is regarded as the fundamental state variable describing the network (Mei et al., 2018; Chizat and Bach, 2018).

Initialization and limiting representation. Assume that the parameters $\{\theta_\alpha(0)\}_{\alpha=1}^m$ are initialized i.i.d. according to a probability measure μ_0 on Ω . Then the empirical measure at initialization converges weakly to μ_0 almost surely as $m \rightarrow \infty$. Consequently, the network output at initialization converges pointwise to the deterministic limit

$$f_{\mu_0}(x) := \int_{\Omega} v \Phi(w^\top x) d\mu_0(v, w).$$

For instance, under i.i.d. Gaussian initialization of the parameters, the limiting initial measure μ_0 is Gaussian on Ω .

Training dynamics and evolving kernel Let $X = (x_1, \dots, x_n) \in \mathbb{R}^{d \times n}$ and $y = (y_1, \dots, y_n) \in \mathbb{R}^n$ denote the training set, and consider the squared loss

$$L(v, W) = \frac{1}{2} \sum_{i=1}^n (f_{v, W}(x_i) - y_i)^2.$$

Under gradient flow, the particle system induces an evolution of the empirical measure $\mu_{m,t}$ in parameter space. More precisely, there is an explicit measure-dependent velocity field

$$b((v, w); \mu) := \begin{pmatrix} \Phi(X^\top w)^\top (y - f_\mu(X)) \\ X \left(\Phi'(X^\top w) \odot (y - f_\mu(X)) \right) v \end{pmatrix},$$

where

$$f_\mu(x) := \int_{\Omega} v \varphi(w^\top x) d\mu(v, w), \quad f_\mu(X) := (f_\mu(x_i))_{i=1}^n,$$

such that the particle dynamics can be written as

$$\dot{\theta}_{t,\alpha} = \frac{\eta}{m} b(\theta_{t,\alpha}; \mu_{m,t}), \quad \alpha = 1, \dots, m.$$

A direct derivation of this system is given in Appendix A.1.3.

Equivalently, the empirical measure solves the continuity equation

$$\partial_t \mu_{m,t} + \nabla_{\theta^\top} \left(\mu_{m,t} \frac{\eta}{m} b(\cdot; \mu_{m,t}) \right) = 0. \quad (2.17)$$

A weak formulation and the associated push-forward representation are given in Appendix A.1.4.

Choosing the mean-field scaling $\eta = m$ yields a nontrivial $O(1)$ evolution in time and, as $m \rightarrow \infty$, one expects $\mu_{m,t} \rightarrow \mu_t$, where the limit μ_t satisfies

$$\partial_t \mu_t + \nabla_{\theta^\top} (\mu_t b(\cdot; \mu_t)) = 0. \quad (2.18)$$

2.2.1 Advantages over the NTK limit.

Recall the finite-width NTK expression (2.8). In the mean-field parameterization $f(x) = \frac{1}{m} \sum_{\alpha=1}^m v_\alpha \varphi(w_\alpha^\top x)$, the empirical tangent kernel is

$$\Theta_t^{(m)}(x, x') = \frac{1}{m^2} \sum_{\alpha=1}^m \left(\varphi(w_{t,\alpha}^\top x) \varphi(w_{t,\alpha}^\top x') + v_{t,\alpha}^2 \varphi'(w_{t,\alpha}^\top x) \varphi'(w_{t,\alpha}^\top x') x^\top x' \right). \quad (2.19)$$

Since $\Theta_t^{(m)} = O(m^{-1})$, the effective operator driving learning is the scaled kernel $\eta \Theta_t^{(m)}$. Under the mean-field scaling $\eta = m$, this quantity has a finite limit:

$$\lim_{m \rightarrow \infty} (\eta \Theta_t^{(m)}(x, x')) = \int_{\Omega} \left(\varphi(w^\top x) \varphi(w^\top x') + v^2 \varphi'(w^\top x) \varphi'(w^\top x') x^\top x' \right) d\mu_t(v, w).$$

As μ_t evolves according to the mean-field transport equation, this limiting kernel also evolves in time. In contrast to the frozen-kernel NTK regime, the mean-field viewpoint therefore captures feature learning through an explicitly time-dependent limiting representation.

Disadvantages. The mean-field scaling yields a deterministic measure-valued evolution $(\mu_t)_{t \geq 0}$ as $m \rightarrow \infty$, but its long-time behavior is generally difficult to characterize. In particular, it is not clear in general whether μ_t converges as $t \rightarrow \infty$, nor which minimizer is selected at the level of measures. Moreover, unlike the NTK limit, extending the above measure-evolution construction beyond two-layer networks is not straightforward, since deeper architectures do not admit the same simple permutation-invariant description in terms of an empirical measure on a fixed parameter space. Several multilayer mean-field-type constructions have been proposed to address this difficulty (Nguyen, 2019; Araújo et al., 2019; Sirignano and Spiliopoulos, 2019; Nguyen and Pham, 2023).

2.3 Spectral bias

A striking and widely reported phenomenon in neural network training is that gradient-based optimization does not fit all components of a target function at the same rate. Instead, networks tend to learn “simple” structure first and refine fine-scale or highly oscillatory structure later. This preference is commonly referred to as *spectral bias* (or the *frequency principle*). In its classical formulation, spectral bias asserts that, when a target function is decomposed into Fourier modes, lower-frequency components are learned earlier and at a faster rate than higher-frequency components (Rahaman et al., 2019).

In this section we briefly review empirical evidence for spectral bias and summarize theoretical viewpoints that relate it to the spectrum of the operator governing training dynamics in the kernel (NTK) regime. We also highlight extensions that study how the phenomenon depends on the input distribution and how it manifests outside the training set.

2.3.1 Empirical evidence and experimental protocols

A standard way to empirically probe spectral bias is to choose a target with a controlled frequency decomposition and to track the frequency content of the network’s prediction f_t during training.

Synthetic Fourier regression. A canonical experiment is 1D regression on a target constructed as a sum of sinusoids,

$$y(z) = \sum_{i=1}^r A_i \sin(2\pi k_i z + \phi_i), \quad z \in [0, 1],$$

and monitoring the discrete Fourier spectrum of the learned predictor as a function of training time. Rahaman et al. (2019) report that, for deep ReLU networks trained by (full-batch) gradient descent, Fourier coefficients at smaller frequencies k_i grow substantially earlier than those at larger k_i , even when the amplitudes are matched or when high-frequency components have larger amplitude (Rahaman et al., 2019). This “frequency-dependent learning speed” is one of the most direct empirical signatures of spectral bias.

Robustness and perturbation tests. A complementary protocol is to train to near-zero training error and then apply random perturbations in parameter space, comparing how the Fourier spectrum of the realized function changes. Rahaman et al. (2019) observe that lower-frequency components of the learned function are substantially more robust to such perturbations than higher-frequency components, suggesting that the network parameterization itself represents low frequencies in a more stable manner (Rahaman et al., 2019).

Distributional effects and “local” frequency learning. Spectral bias is often presented under uniformly sampled inputs, but empirical studies also examine how it interacts with the input distribution. For example, Basri et al. (2020) compare learning under uniform and non-uniform sampling densities and show that the ordering “low frequencies before high frequencies” can be modulated by where samples concentrate: dense regions of the input

space may exhibit faster learning of higher-frequency structure than sparse regions (Basri et al., 2020). This line of experimentation motivates viewing spectral bias as a property of an operator defined jointly by the model and the data distribution, rather than as a purely architectural effect.

2.3.2 Operator spectrum viewpoint and the NTK regime

A common theoretical explanation of spectral bias proceeds by identifying an operator whose eigendecomposition controls learning rates. In the kernel/NTK regime, training dynamics are approximately linear in function space and decompose along eigendirections of the corresponding kernel operator. In this setting, spectral bias can be interpreted more generally as a bias toward learning the leading eigenfunctions of the kernel (rather than specifically Fourier modes) (Bowman and Montufar, 2022).

Decomposition along NTK eigenfunctions. Cao et al. (2020) give a rigorous account of this mechanism in the NTK regime: the training process can be decomposed along eigenfunctions of the NTK operator, with each component converging at a rate determined by the associated eigenvalue. As a consequence, components aligned with larger eigenvalues are learned faster, yielding a principled notion of “spectral preference” (Cao et al., 2020).

In general, the eigenfunctions of the NTK operator depend jointly on the input distribution and the kernel, so they need not coincide with the standard Fourier basis. A simplification occurs in the *infinite-width* (kernel) limit, where the empirical NTK concentrates around a deterministic limit $\bar{\Theta}$ given by an expectation over the random initialization.

To illustrate how Fourier modes arise, consider inputs on the circle S^1 , parameterized by $\theta \in [0, 2\pi)$ and embedded as $x(\theta) = (\cos \theta, \sin \theta) \in \mathbb{R}^2$. Assume isotropic initialization (e.g. Gaussian), so that $R^\top w$ has the same distribution as w for every orthogonal matrix R . Writing $\bar{\Theta}$ as an expectation (cf. Eq. (2.9)) and using this rotational invariance in distribution implies

$$\bar{\Theta}(Rx, Rx') = \bar{\Theta}(x, x') \quad \text{for all orthogonal } R.$$

Since $x(\theta + \alpha) = R_\alpha x(\theta)$ and $x(\theta)^\top x(\theta') = \cos(\theta - \theta')$, it follows that $\bar{\Theta}(\theta, \theta')$ depends only on $\theta - \theta'$; equivalently, there exists κ such that

$$\bar{\Theta}(\theta, \theta') = \kappa(\theta - \theta').$$

Convolution form on S^1 . Let ρ be the uniform probability measure on S^1 , $d\rho(\theta') = d\theta'/(2\pi)$, and define the integral operator

$$(T_{\bar{\Theta}}g)(\theta) := \int_0^{2\pi} \bar{\Theta}(\theta, \theta') g(\theta') \frac{d\theta'}{2\pi}.$$

Using $\bar{\Theta}(\theta, \theta') = \kappa(\theta - \theta')$ and the change of variables $u = \theta - \theta'$, we obtain

$$(T_{\bar{\Theta}}g)(\theta) = \int_0^{2\pi} \kappa(u) g(\theta - u) \frac{du}{2\pi} =: (\kappa * g)(\theta),$$

which is the (circular) convolution operator on S^1 .

Fourier eigenfunctions and frequency ordering. Because $T_{\bar{\Theta}}$ is a convolution, the Fourier modes $\phi_q(\theta) = e^{iq\theta}$, $q \in \mathbb{Z}$, are the eigenfunctions:

$$(T_{\bar{\Theta}}\phi_q)(\theta) = e^{iq\theta} \int_0^{2\pi} \kappa(u) e^{-iqu} \frac{du}{2\pi} = \lambda_q \phi_q(\theta), \quad \lambda_q = \int_0^{2\pi} \kappa(u) e^{-iqu} \frac{du}{2\pi}.$$

If κ is real and even (as in $\kappa(u) = \psi(\cos u)$), then $\lambda_q \in \mathbb{R}$ and $\lambda_q = \lambda_{-q}$, and one may equivalently use the real basis $\{\cos(q\theta), \sin(q\theta)\}$. In kernel gradient dynamics, each mode contracts independently at a rate set by λ_q (e.g. $(1 - \eta\lambda_q)^t$ in discrete time), so modes with larger eigenvalues are learned faster. In symmetric NTK settings on the circle/sphere, the eigenvalues decrease with frequency, leading to the characteristic “low frequencies first” behavior (Basri et al., 2020; Rahaman et al., 2019).

2.3.3 Uniform vs. non-uniform sampling

The Fourier-mode picture above relies not only on a shift-invariant kernel but also on the *uniform* input distribution on S^1 . Under uniform sampling, $d\rho(\theta') = d\theta'/(2\pi)$, shift invariance $\bar{\Theta}(\theta, \theta') = \kappa(\theta - \theta')$ implies that the integral operator is a circular convolution and therefore diagonalizes in the Fourier basis. In this setting, spectral bias can be stated directly in terms of frequency: learning rates are controlled by the eigenvalues λ_q associated with Fourier modes.

Under *non-uniform* sampling, this simplification breaks. If inputs are distributed according to a density $p(\theta)$ on S^1 , then the relevant population operator becomes

$$(T_p g)(\theta) := \int_0^{2\pi} \kappa(\theta - \theta') g(\theta') p(\theta') \frac{d\theta'}{2\pi}.$$

Even when κ depends only on $\theta - \theta'$, the extra factor $p(\theta')$ destroys shift invariance, so T_p is no longer a pure convolution operator and its eigenfunctions need not be the global Fourier modes. As a result, a Fourier harmonic target can project onto multiple eigendirections of T_p , and “frequency” becomes distribution-dependent: the modes learned early are those aligned with the *top eigenfunctions of the density-weighted operator* rather than the lowest Fourier frequencies.

This dependence on p is analyzed explicitly by Basri et al. (2020), who study the kernel regime on the circle/sphere and show that non-uniform densities can produce eigenfunctions with localized oscillatory structure and learning behavior that varies across the input space, with denser regions exhibiting effectively faster learning of finer-scale structure (Basri et al., 2020).

2.3.4 Spectral bias beyond the training set

Most early demonstrations of spectral bias are reported on the training set via Fourier spectra or projections onto empirical kernel eigenvectors. A natural question is whether a comparable phenomenon holds in function space more globally. Bowman and Montufar (2022) address this in the kernel regime by proving bounds that compare finite-width training trajectories to idealized kernel dynamics and conclude that networks inherit the bias of the

NTK integral operator over the input space, not merely at the training samples. In particular, they argue that eigenfunctions of the NTK integral operator are learned at rates corresponding to their eigenvalues, providing a mechanism for spectral bias that persists outside the training set (Bowman and Montufar, 2022).

2.3.5 Discussion and connection to feature learning

The operator-spectrum viewpoint provides a clean account of spectral bias in kernelized regimes, where learning rates are dictated by the spectrum of a fixed (or nearly fixed) operator. Outside the NTK regime, however, features can evolve substantially during optimization, and the relevant operator (e.g. the tangent kernel) may drift. From this perspective, spectral bias can be studied both as a *property of the initial linearized dynamics* and as a phenomenon that may interact with feature learning through time-dependent spectral structure. This interaction motivates empirical analyses that track mode-wise error decay alongside kernel evolution and provides a natural link between kernel-based and mean-field viewpoints.

2.4 Research gap

The preceding discussion shows that spectral bias is well understood in idealized kernel settings. In the infinite-width NTK regime, training is governed by a fixed operator whose spectrum determines the decay rates of the corresponding components of the residual (Jacot et al., 2020; Rahaman et al., 2019; Cao et al., 2020). On S^1 with the uniform measure, this operator becomes a convolution and is diagonalized by the Fourier basis. There is also substantial work on kernel convergence, finite-width corrections, and kernel evolution beyond the strict NTK regime (Golikov et al., 2022; Bordelon and Pehlevan, 2023; Vyas et al., 2022).

What is still missing is a perturbative analysis carried out within a single setting in which the continuum operator, its spectrum, and its eigenfunctions are all explicit. In particular, the existing theory does not yet provide a unified account of how the Fourier description on S^1 is altered first by replacing the continuum measure with finitely many uniform samples, and then by replacing the infinite-width kernel with the kernel induced by a finite-width network.

This thesis studies spectral bias on S^1 by comparing the residual dynamics induced by the continuum infinite-width NTK operator with those induced by finite-sample and finite-width perturbations. Since the continuum operator acts on an infinite-dimensional function space, asking whether the entire Fourier decomposition persists under perturbations is too strong for the scope of this thesis. The high-frequency part of the spectrum is also the most delicate because the eigenvalues become smaller, the gaps between eigenvalues can shrink, and finite samples or finite-width fluctuations have a comparatively larger effect on these modes. We therefore study the question locally in frequency, by fixing a low-frequency cut-off K and analyzing the corresponding truncated Fourier subspace. Throughout the thesis, these comparisons are made after projection onto a fixed truncated Fourier subspace $\mathcal{H}_K$, consisting of the first K Fourier frequencies.

Main research question. To what extent does the spectral decomposition of the continuum infinite-width NTK operator on S^1 persist on a fixed low-frequency Fourier subspace $\mathcal{H}_K$ under finite-sample and finite-width perturbations?

Subquestions.

- (Q1) **Finite-sample perturbation of the continuum operator.** In the infinite-width regime, how does replacing the uniform measure on S^1 by n i.i.d. sampled points affect the action of the continuum NTK operator on the fixed truncated Fourier subspace $\mathcal{H}_K$? More precisely, how close is the sampled operator to the diagonal Fourier prediction on this subspace, and what are the consequences for the associated residual dynamics?
- (Q2) **Finite-width perturbation in the continuum setting.** With the continuum measure on S^1 kept fixed, how does finite network width perturb the frozen tangent operator on the fixed truncated Fourier subspace $\mathcal{H}_K$? More precisely, how close is the projected finite-width operator $P_K T_0^{(m)} P_K$ to the continuum Fourier block $P_K T_\infty P_K$, and how does this perturbation affect the alignment between finite-width eigenspaces and the corresponding Fourier frequency subspaces?

2.5 Contributions

- We derive the infinite-width reference model for the specific network studied in this thesis. Closely related two-layer analyses often simplify the NTK by training only the first-layer weights and biases while keeping the output weights fixed (Ronen et al., 2019; Basri et al., 2020). Here, we keep the output weights trainable and include their contribution to the NTK. This gives an explicit kernel on S^1 , explicit Fourier learning rates, and the baseline used in the rest of the thesis.
- We study how this Fourier-frequency picture changes when the continuum input distribution is replaced by finitely many sampled points. For a fixed low-frequency subspace, we prove non-asymptotic bounds showing that the sampled Fourier structure remains close to the idealized continuum prediction when the sample size is large enough.
- We support the finite-sample analysis with numerical experiments. These experiments compare the predicted frequency-wise learning rates with the rates observed on sampled data, and show that the agreement improves as the number of samples increases.
- We study the effect of finite network width when the tangent kernel is frozen at initialization. On the same low-frequency subspace, we prove that the finite-width operator concentrates around the infinite-width prediction as width increases, and we verify this numerically using eigenspace-alignment diagnostics.

Together, these results show how the clean Fourier description from the infinite-width model changes under finite data and frozen finite width. The fully evolving finite-width tangent kernel during training remains the main direction left open by the thesis.

Chapter 3

Methodology

This chapter defines the operators, projections, and diagnostics used throughout the thesis. The goal is to make clear what each object measures and how it is used in the finite-sample and finite-width experiments. Detailed derivations are given in the appendices

3.1 Mathematical setting

3.1.1 Domain and measure

We identify the unit circle S^1 with the angular domain $\Omega = [0, 2\pi)$, equipped with the uniform probability measure

$$d\mu(\varphi) = \frac{d\varphi}{2\pi}. \quad (3.1)$$

The embedding into $\mathbb{R}^2$ is given by

$$x(\varphi) = (\cos \varphi, \sin \varphi) \in S^1 \subset \mathbb{R}^2. \quad (3.2)$$

This parametrization makes the rotational symmetry of the problem explicit and provides the natural Fourier basis for $L^2(\mu)$.

3.1.2 Network model

We consider a one-hidden-layer ReLU network in NTK parameterization. For inputs $x \in S^1 \subset \mathbb{R}^2$,

$$f_m(x; \theta) = \frac{1}{\sqrt{m}} \sum_{i=1}^m a_i \sigma(w_i^\top x + b_i), \quad (3.3)$$

where $\theta = \{(a_i, w_i, b_i)\}_{i=1}^m$, $\sigma(t) = t_+ = \max\{t, 0\}$, and the parameters are initialized according to

$$a_i \sim \mathcal{N}(0, 1), \quad w_i \sim \mathcal{N}(0, I_2), \quad b_i(0) = 0, \quad (3.4)$$

independently across $i = 1, \dots, m$. The bias parameters are initialized at zero but remain trainable throughout.

The analysis is organized around three related operators:

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1. the infinite-width limiting tangent operator T_∞ ,
2. the finite-width frozen operator $T_0^{(m)}$ at initialization,
3. the exact time-dependent finite-width operator $T_t^{(m)}$ during training.

The limiting operator T_∞ provides the continuum Fourier reference model. The finite-sample analysis asks how this structure is affected by replacing the measure μ with finitely many sampled points. The finite-width analysis asks how the operator generated by finitely many neurons differs from the infinite-width reference, both in the frozen and in the evolving regimes.

3.1.3 Least-squares loss and residual dynamics

Let $y \in L^2(\mu)$ be the target function, and write

$$f_t(\varphi) := f_m(x(\varphi); \theta(t))$$

for the network output at training time t . We consider the least-squares loss

$$\mathcal{L}(\theta) = \frac{1}{2} \|f_m(\cdot; \theta) - y\|_{L^2(\mu)}^2 = \frac{1}{2} \int_0^{2\pi} (f_m(\varphi; \theta) - y(\varphi))^2 d\mu(\varphi). \quad (3.5)$$

We define the residual

$$r_t(\varphi) = f_t(\varphi) - y(\varphi). \quad (3.6)$$

At finite width, the empirical neural tangent kernel at time t is

$$\Theta_t^{(m)}(\varphi, \psi) = \langle \nabla_{\theta} f_m(\varphi; \theta(t)), \nabla_{\theta} f_m(\psi; \theta(t)) \rangle, \quad (3.7)$$

and the corresponding integral operator $T_t^{(m)} : L^2(\mu) \rightarrow L^2(\mu)$ is

$$(T_t^{(m)} f)(\varphi) = \int_0^{2\pi} \Theta_t^{(m)}(\varphi, \psi) f(\psi) d\mu(\psi). \quad (3.8)$$

Under gradient flow, the residual satisfies the exact evolution equation

$$\partial_t r_t = -T_t^{(m)} r_t. \quad (3.9)$$

This equation contains both the frozen and evolving regimes. In the frozen regime, $T_t^{(m)}$ is replaced by its initial value

$$T_0^{(m)} := T_t^{(m)}|_{t=0}, \quad (3.10)$$

whereas in the evolving regime the full time dependence of $T_t^{(m)}$ is retained.

3.2 Continuum reference model

3.2.1 Infinite-width tangent operator

The first object studied in the thesis is the infinite-width limiting tangent operator

$$T_\infty := \lim_{m \rightarrow \infty} T_0^{(m)}, \quad (3.11)$$

whose kernel is denoted by Θ . Under the initialization (3.4), the limiting kernel exists and is deterministic. The explicit formula for Θ and the corresponding Fourier eigenvalues are stated in Chapter 4.4.1 and derived in Appendix B.

3.2.2 Translation invariance and Fourier diagonalization

Because the input distribution is uniform and the limiting kernel is rotation-invariant, $\Theta(\varphi, \psi)$ depends only on the angular difference. The limiting operator therefore takes the convolution form

$$(T_\infty f)(\varphi) = \int_0^{2\pi} \Theta(\varphi - \psi) f(\psi) d\mu(\psi), \quad (3.12)$$

and is diagonalized by the real Fourier basis of $L^2(\mu)$. Its eigenvalues determine the decay rates of the Fourier components of the residual in the ideal fixed-kernel continuum model.

This continuum Fourier decomposition serves as the reference point for the finite-sample and finite-width analyses below.

3.2.3 Real Fourier basis and truncated subspaces

We define the real Fourier basis of $L^2(\mu)$ by

$$\phi_0(\varphi) = 1, \quad (3.13)$$

and, for each $k \geq 1$,

$$\phi_{k,c}(\varphi) = \sqrt{2} \cos(k\varphi), \quad \phi_{k,s}(\varphi) = \sqrt{2} \sin(k\varphi). \quad (3.14)$$

For a fixed cutoff $K \geq 1$, we consider the truncated subspace

$$\mathcal{H}_K = \text{span}\{\phi_0, \phi_{k,c}, \phi_{k,s} : 1 \leq k \leq K\}, \quad (3.15)$$

with dimension

$$d = 2K + 1. \quad (3.16)$$

Let

$$P_K : L^2(\mu) \rightarrow \mathcal{H}_K \quad (3.17)$$

denote the orthogonal projection.

The truncation to $\mathcal{H}_K$ is used throughout the thesis to reduce the continuum problem to a finite-dimensional low-frequency block that can be compared across the continuum, sampled, frozen finite-width, and evolving finite-width settings.

3.3 Finite-sample methodology

3.3.1 Finite-sample question

We first study the effect of replacing the measure μ by finitely many sampled points. Let

$$\varphi_1, \dots, \varphi_n \stackrel{\text{iid}}{\sim} \text{Unif}[0, 2\pi). \quad (3.18)$$

The main question is whether the diagonal Fourier decomposition of the continuum operator remains approximately valid after this replacement.

3.3.2 Sampled objects

To study the sampled operator on $\mathcal{H}_K$, we introduce the feature matrix $\Phi \in \mathbb{R}^{n \times d}$,

$$\Phi_{i,p} = \phi_p(\varphi_i), \quad 1 \leq i \leq n, \quad 1 \leq p \leq d, \quad (3.19)$$

and the normalized sampled kernel matrix

$$A_{ij} = \frac{1}{n} \Theta(\varphi_i - \varphi_j). \quad (3.20)$$

From these we form the Gram matrix

$$G = \frac{1}{n} \Phi^\top \Phi, \quad (3.21)$$

and the compressed operator matrix

$$H = \frac{1}{n} \Phi^\top A \Phi. \quad (3.22)$$

We also introduce the diagonal matrix

$$\Lambda^{(n)} = \text{diag}(\lambda_p^{(n)}), \quad (3.23)$$

whose entries are the finite-sample corrected Fourier eigenvalues defined in Chapter 4.4.1.

3.3.3 Quantities to be controlled

The finite-sample analysis is based on three comparisons:

$$G - I_d, \quad H - \Lambda^{(n)}, \quad A\Phi - \Phi\Lambda^{(n)}. \quad (3.24)$$

The first comparison measures how far the sampled Fourier basis is from being orthonormal. The second measures how far the sampled operator is from the diagonal Fourier prediction on $\mathcal{H}_K$. The third measures how far the sampled Fourier modes are from being eigenvectors of the sampled operator. Together, these are the quantities needed to answer Q1.

3.4 Fourier-block preconditioning methodology

The finite-sample analysis gives a diagonal reference for the sampled operator on a fixed Fourier block. We use this reference to define a diagnostic preconditioning experiment. The goal is to test whether the predicted Fourier eigenvalues explain the observed rate separation between retained Fourier modes.

Fix a retained Fourier block

$$\mathcal{H}_K = \text{span}\{\phi_0, \phi_{1,c}, \phi_{1,s}, \dots, \phi_{K,c}, \phi_{K,s}\},$$

with dimension $d = 2K + 1$. Let

$$\Phi \in \mathbb{R}^{n \times d}$$

be the sampled Fourier feature matrix, and let $A \in \mathbb{R}^{n \times n}$ be the normalized sampled kernel operator. Recall that

$$G = \frac{1}{n} \Phi^\top \Phi, \quad H = \frac{1}{n} \Phi^\top A \Phi. \quad (3.25)$$

The matrix G is the sampled Fourier Gram matrix, while H records the action of the sampled operator A on the retained Fourier basis. When G is invertible, define

$$C_n^{(K)} := G^{-1} H. \quad (3.26)$$

This matrix describes what the sampled operator A does to the Fourier coefficients on the retained block. Indeed, if $u = \Phi z$, then z is the coefficient vector of u in the sampled Fourier basis. After applying A , the least-squares coefficients of Au in the same sampled Fourier basis are

$$G^{-1} \frac{1}{n} \Phi^\top (Au) = G^{-1} \frac{1}{n} \Phi^\top A \Phi z = C_n^{(K)} z. \quad (3.27)$$

Thus $C_n^{(K)}$ is the object that should be compared with the diagonal finite-sample prediction

$$\Lambda^{(n)} = \text{diag}(\lambda_p^{(n)}). \quad (3.28)$$

The finite-sample theory predicts that, on the retained block,

$$C_n^{(K)} \approx \Lambda^{(n)}. \quad (3.29)$$

This means that, on the retained Fourier block, the sampled operator is expected to act mainly by rescaling each Fourier coefficient by its predicted eigenvalue, with only small mixing between different retained Fourier modes.

We now define two preconditioners from this comparison. First note that, for any sampled vector $v \in \mathbb{R}^n$,

$$G^{-1} \frac{1}{n} \Phi^\top v$$

gives the least-squares Fourier coefficients of the projection of v onto the sampled span of the retained modes. Thus, to correct the action of A on the retained block, we first apply

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A , then project back onto the retained Fourier span, and finally rescale the resulting Fourier coefficients.

The theory preconditioner uses the diagonal prediction $\Lambda^{(n)}$:

$$P_{\text{th}} := \frac{1}{n} \Phi (\Lambda^{(n)})^{-1} G^{-1} \Phi^\top. \quad (3.30)$$

This operator projects a sampled vector onto the retained Fourier span and rescales each retained Fourier coefficient by the inverse predicted eigenvalue. It uses the analytic prediction $\Lambda^{(n)}$, and therefore does not use the observed matrix $C_n^{(K)}$.

The empirical preconditioner uses the observed retained block:

$$P_{\text{emp}} := \frac{1}{n} \Phi (C_n^{(K)})^{-1} G^{-1} \Phi^\top. \quad (3.31)$$

This operator uses the actual sampled matrix $C_n^{(K)}$. It therefore corrects both the diagonal scaling and the mixing between retained Fourier modes that is present for the particular sampled points.

We compare three sampled dynamics:

$$B_{\text{base}} := A, \quad B_{\text{th}} := P_{\text{th}} A, \quad B_{\text{emp}} := P_{\text{emp}} A. \quad (3.32)$$

The baseline B_{base} is the original sampled kernel operator. The operators B_{th} and B_{emp} apply the sampled kernel operator and then correct its action through the retained Fourier block.

For sampled predictions $f_t \in \mathbb{R}^n$, targets $y \in \mathbb{R}^n$, and residuals

$$r_t = f_t - y,$$

we compare the gradient-flow dynamics

$$\frac{df_t}{dt} = -B r_t, \quad \frac{dr_t}{dt} = -B r_t, \quad (3.33)$$

with

$$B \in \{B_{\text{base}}, B_{\text{th}}, B_{\text{emp}}\}.$$

The effect is easiest to see on a vector $u = \Phi z$ in the retained Fourier span. Applying A and projecting back to the retained block gives the coefficient vector $C_n^{(K)} z$. Therefore

$$B_{\text{th}} \Phi z = \Phi (\Lambda^{(n)})^{-1} C_n^{(K)} z, \quad (3.34)$$

while

$$B_{\text{emp}} \Phi z = \Phi (C_n^{(K)})^{-1} C_n^{(K)} z = \Phi z. \quad (3.35)$$

Thus the empirical preconditioner exactly flattens the observed retained block, provided $C_n^{(K)}$ is invertible. The theory preconditioner does the same only when the observed block is close to the diagonal prediction $\Lambda^{(n)}$.

In the numerical experiments, ill-conditioned inverses are replaced by small ridge-regularized inverses when needed, for example $(\Lambda^{(n)} + \tau I)^{-1}$ or $(C_n^{(K)} + \tau I)^{-1}$ with $\tau > 0$. The purpose of this comparison is diagnostic. If B_{th} and B_{emp} behave similarly, then the predicted Fourier eigenvalues explain most of the rate differences on the retained block. If B_{emp} is noticeably better, then the sampled points introduce mixing between retained Fourier modes that is not captured by the diagonal prediction alone.

3.5 Finite-width frozen-kernel methodology

3.5.1 Finite-width question

We next remove sampling randomness and return to the continuum input space S^1 , but keep the network width finite. The question is then whether the low-frequency block of the frozen finite-width tangent operator remains close to the continuum Fourier block predicted by the infinite-width theory.

3.5.2 One-neuron tangent features and operators

For each neuron r , define its tangent feature at initialization by

$$\psi_r(\varphi) = \nabla_{(a_r, w_r, b_r)} (a_r \sigma(w_r^\top x(\varphi) + b_r)). \quad (3.36)$$

The corresponding one-neuron kernel contribution is

$$\Theta^{(r)}(\varphi, \psi) = \psi_r(\varphi)^\top \psi_r(\psi), \quad (3.37)$$

and the associated one-neuron operator is

$$(T^{(r)}f)(\varphi) = \int_0^{2\pi} \Theta^{(r)}(\varphi, \psi) f(\psi) d\mu(\psi). \quad (3.38)$$

At initialization, the frozen finite-width tangent operator is the average

$$T_0^{(m)} = \frac{1}{m} \sum_{r=1}^m T^{(r)}. \quad (3.39)$$

3.5.3 Projected frozen operator

Let $P_K : L^2(\mu) \rightarrow \mathcal{H}_K$ denote the orthogonal projection onto the truncated Fourier subspace $\mathcal{H}_K$. The low-frequency block of the frozen finite-width operator is

$$B_m^{(K)} := P_K T_0^{(m)} P_K. \quad (3.40)$$

The corresponding continuum reference block is

$$\Lambda_K := P_K T_\infty P_K = \text{diag}(\lambda_0, \lambda_1, \lambda_1, \dots, \lambda_K, \lambda_K). \quad (3.41)$$

The finite-width frozen analysis asks whether $B_m^{(K)}$ remains close to Λ_K with high probability as m increases.

3.5.4 Matrix representation of the frozen block

Using the ordered basis

$$(\phi_0, \phi_{1,c}, \phi_{1,s}, \dots, \phi_{K,c}, \phi_{K,s}),$$

the projected operator $B_m^{(K)}$ is represented by the matrix with entries

$$(B_m^{(K)})_{pq} = \langle \phi_p, T_0^{(m)} \phi_q \rangle_{L^2(\mu)}. \quad (3.42)$$

Equivalently,

$$(B_m^{(K)})_{pq} = \frac{1}{m} \sum_{r=1}^m \xi_{r,pq}, \quad \xi_{r,pq} := \langle \phi_p, T^{(r)} \phi_q \rangle, \quad (3.43)$$

so that $B_m^{(K)}$ is an empirical average of i.i.d. one-neuron matrices. This is the basic structural fact underlying the finite-width frozen concentration theorem.

3.5.5 Projection-based Fourier-plane alignment

The continuum operator is diagonal in the Fourier basis. For each frequency $k \geq 1$, the cosine and sine modes have the same continuum eigenvalue λ_k . The relevant object is therefore the two-dimensional frequency subspace

$$\mathcal{F}_k := \text{span}\{\phi_{k,c}, \phi_{k,s}\}, \quad k \geq 1,$$

rather than a particular cosine-sine basis inside this subspace. For the constant mode, we set

$$\mathcal{F}_0 := \text{span}\{\phi_0\}.$$

Thus $d_0 = 1$ and $d_k = 2$ for $k \geq 1$.

To test whether the frozen finite-width operator preserves this Fourier structure, we compare each Fourier subspace $\mathcal{F}_k$ with a matched eigenspace of the frozen finite-width operator. The operator is defined on the continuum S^1 , but in the numerical implementation we represent functions on a uniform grid

$$\theta_j = \frac{2\pi j}{N}, \quad j = 0, \dots, N-1.$$

All vectors and matrices in this diagnostic are therefore finite-dimensional representations of continuum functions on this grid.

Let $U_k \in \mathbb{R}^{N \times d_k}$ be an orthonormal basis for the grid representation of $\mathcal{F}_k$, and define the corresponding orthogonal projection

$$P_k := U_k U_k^\top. \quad (3.44)$$

From the eigendecomposition of the frozen finite-width NTK, we build a matched empirical subspace $\widehat{\mathcal{F}}_k$ of the same dimension d_k . We select the d_k empirical eigenvectors with largest overlap with $\mathcal{F}_k$. If v_j is an empirical eigenvector, its overlap with $\mathcal{F}_k$ is measured by

$$\|U_k^\top v_j\|_2^2.$$

If $\widehat{U}_k \in \mathbb{R}^{N \times d_k}$ is an orthonormal basis of the selected empirical subspace $\widehat{\mathcal{F}}_k$, define

$$\widehat{P}_k := \widehat{U}_k \widehat{U}_k^\top. \quad (3.45)$$

We measure the mismatch between the Fourier subspace and the matched empirical subspace using the projector distance

$$\epsilon_{k,F}^{(m)} := \|\widehat{P}_k - P_k\|_F. \quad (3.46)$$

This quantity is invariant to sign changes of eigenvectors and to changes of orthonormal basis inside the subspace. This is necessary for the two-dimensional frequency pairs $k \geq 1$, where rotating the cosine-sine basis inside $\mathcal{F}_k$ should not count as an error.

We also report principal angles between $\widehat{\mathcal{F}}_k$ and $\mathcal{F}_k$. Principal angles are a standard way to compare finite-dimensional subspaces (Björck and Golub, 1973). If U_k and $\widehat{U}_k$ are orthonormal bases for the two subspaces, then the cosines of the principal angles are the singular values of $U_k^\top \widehat{U}_k$. We therefore define

$$0 \leq \theta_{k,1} \leq \dots \leq \theta_{k,d_k} \leq \frac{\pi}{2}$$

by

$$\cos \theta_{k,i} = s_i(U_k^\top \widehat{U}_k), \quad i = 1, \dots, d_k, \quad (3.47)$$

where the singular values are ordered from largest to smallest.

For $k = 0$, there is one principal angle because both spaces are one-dimensional. For $k \geq 1$, there are two principal angles because both spaces are two-dimensional. The first angle measures the best aligned direction between the two planes. The second angle measures the remaining independent direction. Both angles must be small for the full Fourier plane to be well aligned with the matched empirical eigenspace.

The projector distance and the principal angles are related by

$$\|\widehat{P}_k - P_k\|_F^2 = 2 \sum_{i=1}^{d_k} \sin^2 \theta_{k,i}. \quad (3.48)$$

Thus the principal angles give direction-wise information about the alignment, while $\epsilon_{k,F}^{(m)}$ gives one scalar summary of the same subspace mismatch.

3.6 Finite-width evolving-kernel methodology

3.6.1 Projected evolving operator

To study the actual training dynamics, we retain the time dependence of the empirical tangent operator $T_t^{(m)}$. Its low-frequency block is

$$C_t^{(K)} := P_K T_t^{(m)} P_K. \quad (3.49)$$

If $b_t = (I - P_K)r_t$ is the residual component outside the retained Fourier block, then $T_t^{(m)}b_t$ need not remain outside that block. Its projection back into $\mathcal{H}_K$ is

$$L_t^{(K)} b_t = P_K T_t^{(m)} (I - P_K) r_t. \quad (3.50)$$

We therefore define

$$L_t^{(K)} := P_K T_t^{(m)} (I - P_K). \quad (3.51)$$

This operator represents high-to-low frequency coupling. It measures how residual components outside the retained block can influence the evolution of the retained Fourier coefficients after the tangent operator is applied.

3.6.2 Projected residual dynamics

Let

$$a_t := P_K r_t, \quad b_t := (I - P_K) r_t. \quad (3.52)$$

Projecting the exact residual equation (3.9) gives

$$\partial_t a_t = -C_t^{(K)} a_t - L_t^{(K)} b_t. \quad (3.53)$$

Thus the low-frequency dynamics is driven by two terms: the operator restricted to $\mathcal{H}_K$, and the contribution from residual components outside $\mathcal{H}_K$.

3.6.3 Frozen versus evolving comparison

The frozen and evolving regimes differ only through the time dependence of the operator. In the frozen regime,

$$C_t^{(K)} \equiv C_0^{(K)} = B_m^{(K)},$$

whereas in the evolving regime the operator changes during training. This motivates the decomposition

$$C_t^{(K)} = \Lambda_K + (C_0^{(K)} - \Lambda_K) + (C_t^{(K)} - C_0^{(K)}), \quad (3.54)$$

which separates three effects:

1. the ideal infinite-width Fourier block Λ_K ,
2. the finite-width initialization error $C_0^{(K)} - \Lambda_K$,
3. the time-dependent kernel drift $C_t^{(K)} - C_0^{(K)}$.

The frozen-kernel theory studies the second term. The comparison between frozen and evolving training then studies the third term, namely how much the tangent operator changes during training which is largely believed to be the driving factor for feature learning.

3.7 Summary of the methodology

The methodology consists of three comparisons.

First, the continuum infinite-width operator T_∞ gives the ideal Fourier-diagonal reference model on S^1 . In this model, the Fourier modes are exact eigenfunctions and the eigenvalues determine the residual decay rates.

Second, the finite-sample analysis studies what changes when the continuum measure is replaced by finitely many sampled points. The main objects are the sampled Fourier Gram matrix G , the matrix H describing the action of the sampled operator on the retained Fourier basis, and the difference $A\Phi - \Phi\Lambda^{(n)}$ between the sampled operator action and the diagonal Fourier prediction. These objects are used to answer Q1. The preconditioning experiment is a diagnostic for the same question: it tests whether the predicted Fourier eigenvalues explain the observed rate separation on the retained block.

Third, the finite-width analysis studies what changes when the infinite-width operator is replaced by the tangent operator of a finite-width network. The frozen analysis focuses on $B_m^{(K)} = P_K T_0^{(m)} P_K$, the low-frequency block of the tangent operator at initialization. The evolving analysis tracks the time-dependent block $C_t^{(K)} = P_K T_t^{(m)} P_K$ and the high-to-low coupling $L_t^{(K)} = P_K T_t^{(m)} (I - P_K)$. The projector-error and principal-angle diagnostics compare the Fourier frequency subspaces with the matched eigenspaces of the frozen finite-width operator. Together, these objects set up the finite-width part of Q2.

Chapter 4

Results

4.1 Continuum kernel on S^1

We first state the explicit form of the limiting neural tangent kernel on the circle and the corresponding Fourier eigenvalues of the integral operator T introduced in Chapter 3. The calculation uses the standard infinite-width NTK description of Jacot et al. (2020), together with the arc-cosine formulas for ReLU Gaussian averages (Cho and Saul, 2009). Related spectral calculations for ReLU NTKs on the sphere appear in Ronen et al. (2019); Bietti and Mairal (2019); Cao et al. (2020). The derivation in the assumptions used here is deferred to Appendix B.

Proposition 4.1 (Limiting NTK on S^1). *Let Θ be the limiting NTK associated with the model (3.3)–(3.4). If $\delta \in [0, \pi]$ denotes the angular distance between two points on S^1 , then Θ depends only on δ and is given by*

$$\Theta(\delta) = \frac{1}{2\pi} \left(\sin \delta + 2(\pi - \delta) \cos \delta + (\pi - \delta) \right). \quad (4.1)$$

Consequently, the operator T takes the convolution form

$$(Tf)(\varphi) = \int_0^{2\pi} \Theta(\varphi - \psi) f(\psi) d\mu(\psi). \quad (4.2)$$

where $\Theta(\varphi - \psi)$ denotes the even 2π -periodic extension of $\Theta(\delta)$.

Since T is a convolution operator, it is diagonalized by the real Fourier basis introduced in (3.13)–(3.14).

Theorem 4.2 (Fourier eigenvalues of the limiting NTK). *Let λ_k denote the eigenvalue of T corresponding to the frequency k . Then*

$$\lambda_0 = \frac{1}{4} + \frac{3}{\pi^2}, \quad \lambda_1 = \frac{1}{4} + \frac{1}{\pi^2}, \quad (4.3)$$

and for $k \geq 2$,

$$\lambda_k = \begin{cases} \frac{k^2 + 3}{\pi^2(k^2 - 1)^2}, & k \geq 2 \text{ even}, \\ \frac{1}{\pi^2 k^2}, & k \geq 3 \text{ odd}. \end{cases} \quad (4.4)$$

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In particular, the decay rates of the Fourier components of the residual are determined by these eigenvalues.

Corollary 4.3 (Spectrum without the trainable bias). *Let $\Theta^{(\text{nb})}$ denote the limiting NTK obtained by removing the trainable hidden-bias contribution, while keeping the output-weight and input-weight contributions. Then, on S^1 ,*

$$\Theta^{(\text{nb})}(\delta) = \frac{1}{2\pi} \left(\sin \delta + 2(\pi - \delta) \cos \delta \right). \quad (4.5)$$

Let $\lambda_k^{(\text{nb})}$ denote the corresponding Fourier eigenvalue. Then

$$\lambda_0^{(\text{nb})} = \frac{3}{\pi^2}, \quad \lambda_1^{(\text{nb})} = \frac{1}{4}, \quad (4.6)$$

and, for $k \geq 2$,

$$\lambda_k^{(\text{nb})} = \begin{cases} \frac{k^2 + 3}{\pi^2(k^2 - 1)^2}, & k \geq 2 \text{ even}, \\ 0, & k \geq 3 \text{ odd}. \end{cases} \quad (4.7)$$

Consequently, comparing with Theorem 4.2, the trainable bias leaves the even frequencies $k \geq 2$ unchanged, increases the constant and $k = 1$ eigenvalues, and supplies the nonzero eigenvalues of all odd modes $k \geq 3$. Thus the odd modes $k \geq 3$ are absent from the no-bias fixed-kernel dynamics but present in the model studied here.

4.1.1 Continuum fixed-kernel dynamics

The eigenvalues in Theorem 4.2 determine the ideal fixed-kernel dynamics. If

$$r_t = \sum_p \alpha_p(t) \phi_p,$$

then gradient flow gives

$$\alpha_p(t) = \exp(-\lambda_p t) \alpha_p(0). \quad (4.8)$$

The normalized amplitude $|\alpha_p(t)|/|\alpha_p(0)|$ is therefore set by the eigenvalue of the corresponding mode.

Figure 4.1 shows this diagonal continuum dynamics for the kernel with and without the trainable-bias contribution. Low-frequency components decay faster than high-frequency components.

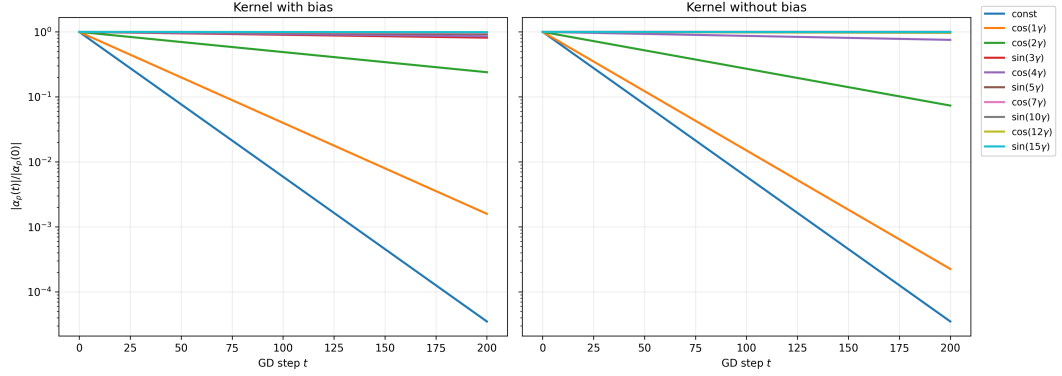


Figure 4.1: Continuum fixed-kernel dynamics for the first 200 gradient descent steps, comparing the kernel with and without the trainable-bias contribution. The curves show normalized mode amplitudes of the residual $|\alpha_p(t)|/|\alpha_p(0)|$ under the diagonal Fourier dynamics.

Figure 4.2 isolates odd Fourier modes. With the bias contribution, these modes have nonzero eigenvalues. Without the bias contribution, the plotted odd modes have zero eigenvalues and remain constant under the fixed-kernel dynamics.

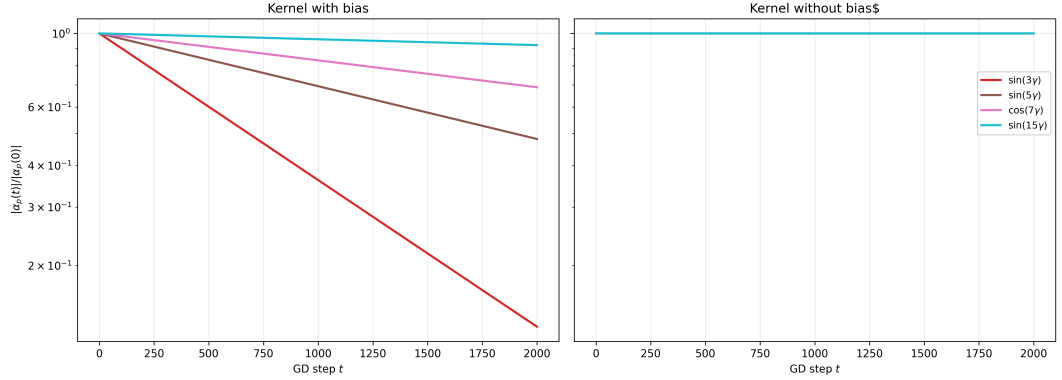


Figure 4.2: Odd-mode continuum fixed-kernel dynamics with and without the trainable-bias contribution. With the bias contribution, the plotted odd modes have nonzero eigenvalues. Without it, the plotted odd modes have zero eigenvalues and therefore remain constant under the fixed-kernel dynamics.

4.2 Finite-sample control on a truncated Fourier subspace

For each basis index p , let $k(p)$ denote the frequency of the corresponding basis function ϕ_p , and write

$$\lambda_p := \lambda_{k(p)}.$$

We now turn to the sampled setting introduced in Chapter 3. Throughout, $K \geq 1$ is fixed, $d = 2K + 1$, and the truncated Fourier subspace $\mathcal{H}_K$ is given by (3.15). The matrices Φ , A , G , and H are defined in (3.19)–(3.22).

Proposition 4.4 (Expectation of the restricted sampled operator). *Let*

$$\lambda_p^{(n)} = \left(1 - \frac{1}{n}\right) \lambda_p + \frac{\Theta(0)}{n}, \quad (4.9)$$

and let

$$\Lambda^{(n)} = \text{diag}(\lambda_p^{(n)}). \quad (4.10)$$

Then

$$\mathbb{E}[H] = \Lambda^{(n)}. \quad (4.11)$$

The next theorem gives explicit concentration bounds for the three sampled quantities that appear in the methodology chapter.

Theorem 4.5 (Finite-sample concentration on the truncated Fourier subspace). *Assume that*

$$\kappa := \sup_{\theta \in [0, 2\pi)} |\Theta(\theta)| < \infty. \quad (4.12)$$

Then, for every $\varepsilon > 0$,

$$\mathbb{P}(\|G - I_d\|_{\max} \geq \varepsilon) \leq d(d+1) \exp\left(-\frac{n\varepsilon^2}{4 + \frac{4}{3}\varepsilon}\right), \quad (4.13)$$

$$\mathbb{P}(\|H - \Lambda^{(n)}\|_{\max} \geq \varepsilon) \leq 2d^2 \exp\left(-\frac{n\varepsilon^2}{32\kappa^2}\right), \quad (4.14)$$

and, for $n \geq 2$,

$$\mathbb{P}(\|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \geq \varepsilon) \leq 2nd \exp\left(-\frac{n\varepsilon^2}{2\kappa^2 + 2\kappa\varepsilon}\right). \quad (4.15)$$

[Draft note for David & Alexander: The following corollary is useful for the bounds plotted below, but it may be too technical for the main Results chapter. I was thinking I could say something like: "Combining the three concentration bounds gives simultaneous high-probability control of $G - I_d$, $H - \Lambda^{(n)}$, and $A\Phi - \Phi\Lambda^{(n)}$. The explicit statement is given in the appendix."]

Corollary 4.6 (High-probability bounds). *Let $\delta \in (0, 1)$, and define*

$$u_{n,d,\delta}^{(G)} := \log\left(\frac{3d(d+1)}{\delta}\right), \quad (4.16)$$

$$u_{n,d,\delta}^{(H)} := \log\left(\frac{6d^2}{\delta}\right), \quad (4.17)$$

and

$$u_{n,d,\delta}^{(A)} := \log\left(\frac{6nd}{\delta}\right). \quad (4.18)$$

Now set

$$\alpha_{n,d,\delta} := 2\sqrt{\frac{u_{n,d,\delta}^{(G)}}{n}} + \frac{2}{3}\frac{u_{n,d,\delta}^{(G)}}{n}, \quad (4.19)$$

$$\beta_{n,d,\delta} := \kappa\sqrt{\frac{32}{n}u_{n,d,\delta}^{(H)}}, \quad (4.20)$$

and

$$\gamma_{n,d,\delta} := \kappa\sqrt{\frac{2}{n}u_{n,d,\delta}^{(A)}} + \kappa\frac{u_{n,d,\delta}^{(A)}}{n}. \quad (4.21)$$

Then, with probability at least $1 - \delta$,

$$\|G - I_d\|_{\max} \leq \alpha_{n,d,\delta}, \quad \|H - \Lambda^{(n)}\|_{\max} \leq \beta_{n,d,\delta}, \quad \|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \leq \gamma_{n,d,\delta}. \quad (4.22)$$

The previous theorem shows that the sampled operator remains close to the continuum Fourier decomposition on the truncated subspace $\mathcal{H}_K$. To express this directly at the level of the induced matrix on coefficients, define

$$C := G^{-1}H, \quad (4.23)$$

whenever G is invertible.

Thus, when the sampled Gram matrix is well conditioned, the coefficient-level restricted operator $C = G^{-1}H$ is close to the diagonal finite-sample prediction $\Lambda^{(n)}$.

4.2.1 Empirical validation of finite-sample Fourier structure

The finite-sample bounds above control three target-independent quantities: the sampled Fourier Gram matrix, the compressed sampled operator on the retained Fourier coordinates, and the sampled eigen-action error. We first evaluate these quantities directly before considering residual dynamics for a particular target.

Figure 4.3 shows the empirical behavior of the three errors appearing in Theorem 4.5:

$$\|G - I_d\|_{\max}, \quad \|H - \Lambda^{(n)}\|_{\max}, \quad \|A\Phi - \Phi\Lambda^{(n)}\|_{\max}.$$

All three errors decrease as n increases. The plotted high-probability envelopes are conservative, but they have the same decreasing trend.

4. RESULTS

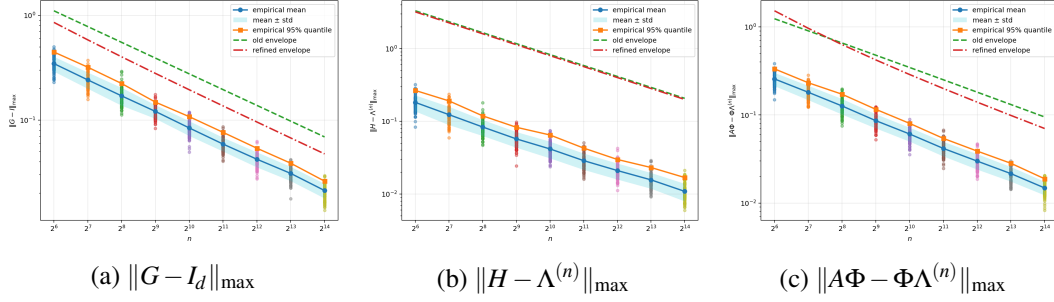


Figure 4.3: Finite-sample diagnostics for the sampled Fourier approximation. The three plotted errors decrease with n . The theoretical envelopes are conservative, but they show the same decreasing trend. [Draft note for David & Alexander: I was trying to use Bernstein's inequality and I managed to improve the bounds slightly, still not as tight but they seem to be better in the first and the third plot than the previous bound I had derived (the green line is the previous bound I came up with and the red is the new one).]

Under random sampling, the continuum mode-decay identity (4.8) is no longer exact, because the sampled Fourier modes are not exact eigenvectors of the sampled operator. The finite-sample theory therefore predicts approximate mode-wise decay.

Figure 4.4 shows the first 200 gradient descent steps of sampled kernel dynamics for $n = 4096$ uniformly sampled points. The solid curves show the observed normalized residual amplitudes, while the dashed curves show the diagonal prediction from $\lambda_p^{(n)}$. The observed curves closely follow the diagonal prediction during early training, especially for the dominant low-frequency modes.

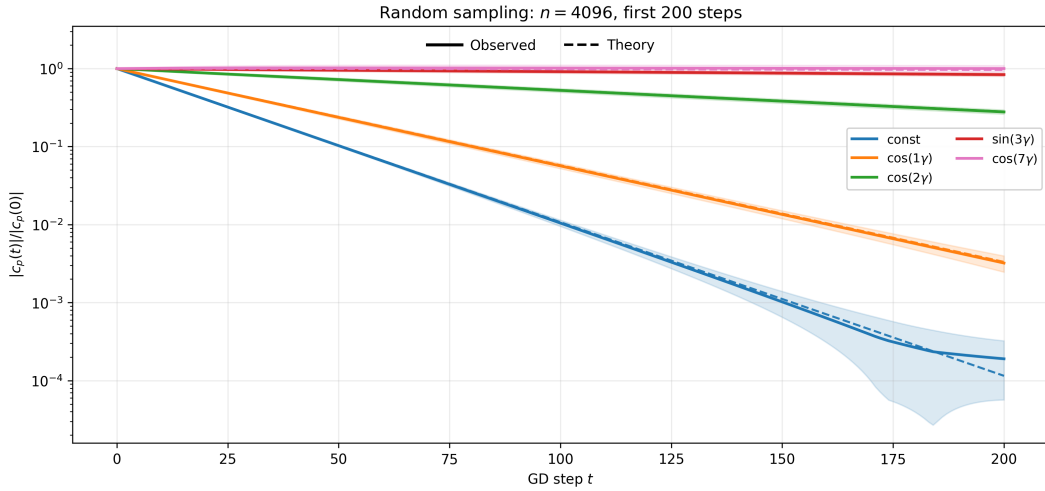


Figure 4.4: Random sampling with $n = 4096$, first 200 gradient descent steps. Solid curves show the observed normalized residual amplitudes $|\alpha_p(t)|/|\alpha_p(0)|$, while dashed curves show the diagonal prediction from the finite-sample corrected eigenvalues $\lambda_p^{(n)}$.

Figure 4.5 shows the higher-frequency retained components over 2000 steps. The initial

decay rates are still well predicted by the diagonal model. At later times, the observed curves deviate from the diagonal prediction and can stop above the ideal prediction.

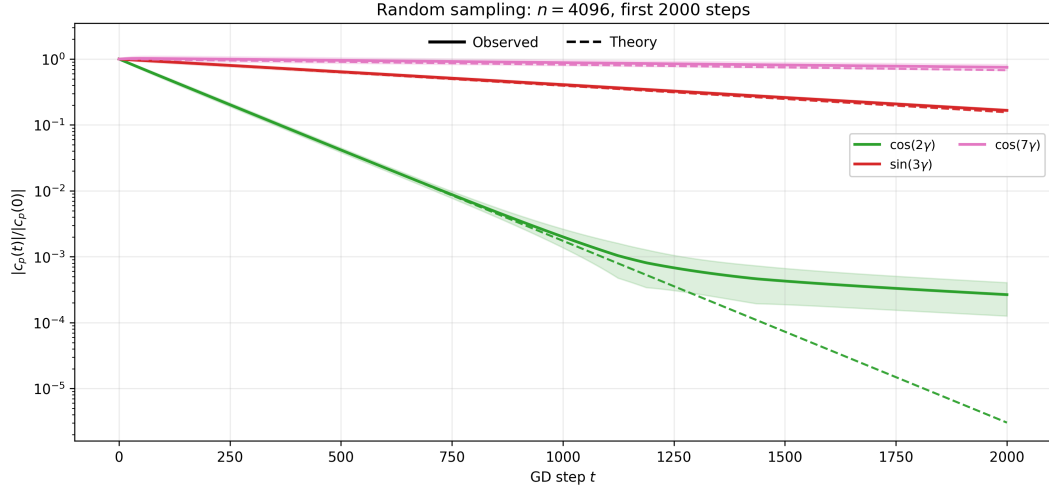


Figure 4.5: Random sampling with $n = 4096$, higher-frequency retained modes over 2000 gradient descent steps. The diagonal prediction captures the initial decay rates, while small deviations from this prediction become visible at later times.

[Draft note for David & Alexander: I wonder if I should also include the following figure which used more samples to show that the deviation in $\cos(2\gamma)$ occur slightly later (the earlier figure with $n=4096$ starts deviating before 10^{-3} but $n=32768$ starts after that point.)]

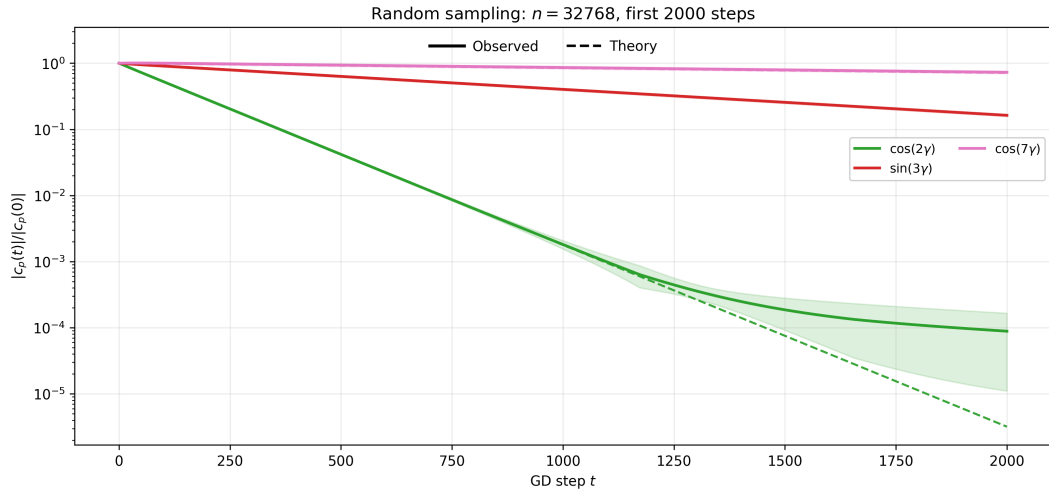


Figure 4.6: Random sampling with $n = 32768$, higher-frequency retained modes over 2000 gradient descent steps. The diagonal prediction captures the initial decay rates, while small deviations from this prediction become visible at later times than $n = 4096$ case.

4.3 Preconditioned sampled dynamics

We next apply the Fourier-block preconditioning construction from Section 3.4. The goal is to compare the baseline sampled dynamics with two block-corrected dynamics: one using the diagonal finite-sample prediction and one using the observed sampled Fourier block.

4.3.1 Experimental setup

The three sampled dynamics are

$$B_{\text{base}} = A, \quad B_{\text{th}} = P_{\text{th}}A, \quad B_{\text{emp}} = P_{\text{emp}}A.$$

Here P_{th} is defined in (3.30) and uses the diagonal prediction $\Lambda^{(n)}$. The empirical preconditioner P_{emp} , defined in (3.31), uses the observed coefficient-space block $C_n^{(K)} = G^{-1}H$.

In this experiment, we use the target

$$f^*(\gamma) = 1 + 0.9\cos(\gamma) + 0.5\cos(2\gamma) - 0.35\sin(3\gamma) + 0.25\cos(7\gamma). \quad (4.24)$$

The retained preconditioning block contains all active target frequencies shown in the residual plots.

4.3.2 Mode-wise residual decay

Figure 4.7 shows the residual amplitudes for the baseline, theory-preconditioned, and empirically preconditioned dynamics. The baseline dynamics shows separated decay rates across retained modes. Both preconditioned dynamics reduce this separation, with the empirical preconditioner giving the flattest curves.

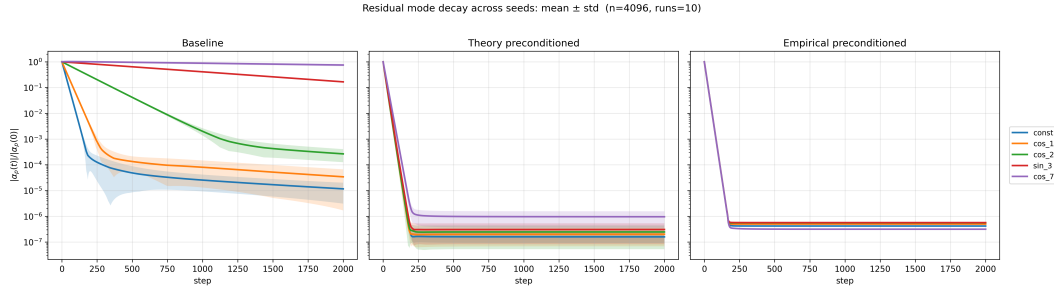


Figure 4.7: Preconditioned sampled dynamics at $n = 4096$. The baseline dynamics shows separated decay rates across retained Fourier modes. Both preconditioned dynamics make the retained residual amplitudes decay on more similar time scales. The empirical preconditioner gives the most uniform decay, while the theory preconditioner uses only the predicted diagonal scaling.

4.3.3 Function-space fit

Figure 4.8 compares the final prediction on a dense probe grid. The baseline fit leaves a visible residual, while both preconditioned flows more closely match the target.

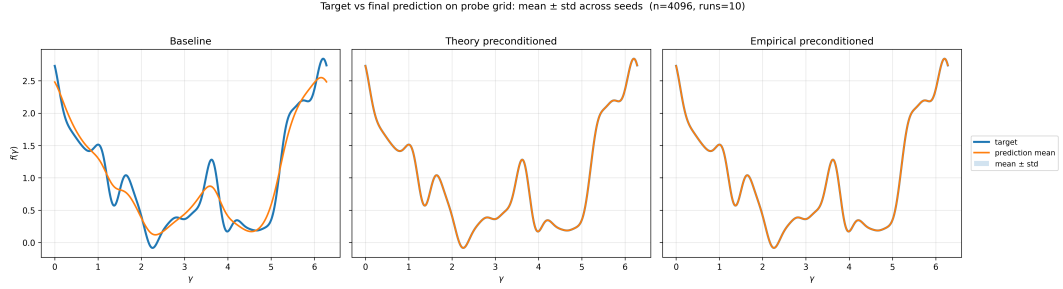


Figure 4.8: Final prediction on a dense probe grid for the same experiment as Figure 4.7. The baseline fit leaves a visible residual, while both preconditioned dynamics closely match the target. The theory and empirical preconditioners are nearly indistinguishable at the level of the final fitted function.

4.3.4 Theory versus empirical preconditioning

Figure 4.9 shows the spectrum of the theory-preconditioned block as n varies. The ideal flattened spectrum is equal to 1. The plotted spectrum becomes flatter as n increases.

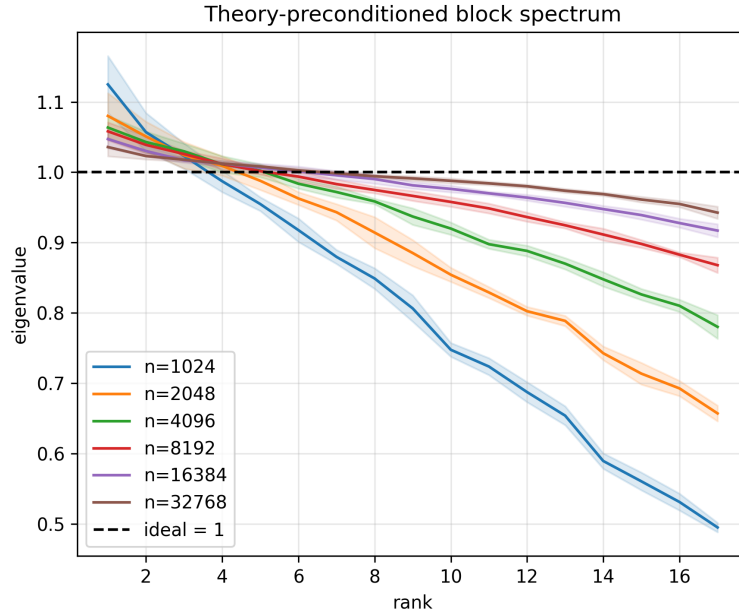


Figure 4.9: Theory-preconditioned block spectrum as a function of sample size. The ideal flattened spectrum is 1. As n increases, the spectrum becomes flatter, showing that the theory preconditioner approaches the empirical block inverse in the large-sample regime.

Figure 4.10 compares the theory and empirical constructions at two levels. The preconditioner P is an operator on sampled function values. Given a vector in $\mathbb{R}^n$, it projects that vector onto the retained Fourier block, rescales the retained Fourier coefficients, and maps

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the result back to $\mathbb{R}^n$. Thus the gap between P_{th} and P_{emp} measures how different the two correction steps are before they are combined with the sampled kernel operator.

The product PA is the operator that is actually used in the residual dynamics. It first applies the sampled kernel operator A to the residual, and then applies the correction P to the retained Fourier components of the result. Thus the gap between B_{th} and B_{emp} measures how different the resulting training dynamics are. Both gaps decrease with sample size.

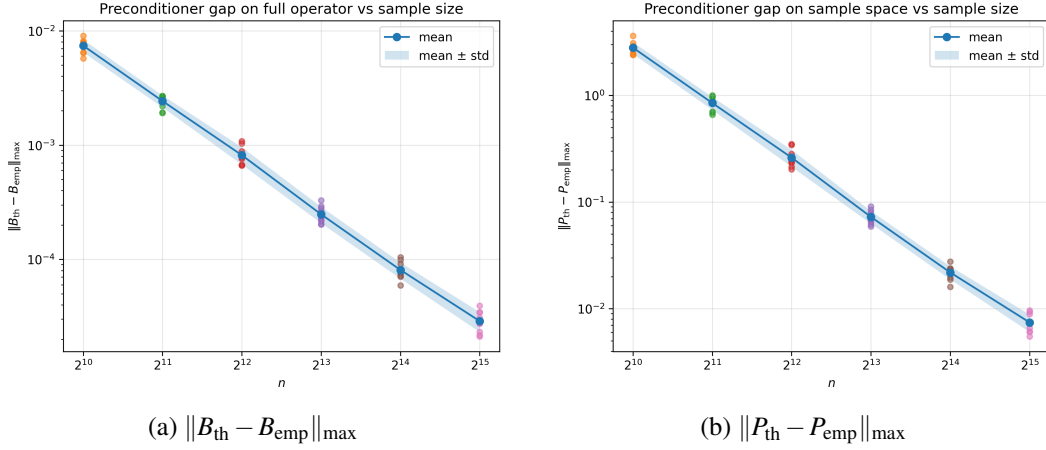


Figure 4.10: Gap between the theory and empirical preconditioners as sample size increases. The empirical preconditioner uses the observed block $C = G^{-1}H$, while the theory preconditioner uses the diagonal approximation $\Lambda^{(n)}$. The decreasing gap is consistent with the finite-sample result $C \approx \Lambda^{(n)}$.

4.4 Finite-width control of the frozen tangent operator

We now turn from sampling error to width-induced error. Throughout this section, the input space remains the continuum S^1 , and the object of interest is the frozen finite-width operator $B_m^{(K)}$ on the truncated Fourier subspace $\mathcal{H}_K$, introduced in (3.40). The corresponding continuum reference block Λ_K was defined in (3.41).

The next theorem shows that, for each fixed K , the frozen finite-width block is a controlled perturbation of the continuum Fourier block.

Theorem 4.7 (Finite-width concentration of the frozen low-mode operator). *Fix $K \geq 1$, and let $d = 2K + 1$. Then*

$$\mathbb{E}[B_m^{(K)}] = \Lambda_K. \quad (4.25)$$

Moreover, there exists a universal constant $c > 0$ such that, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\|B_m^{(K)} - \Lambda_K\|_{\max} \geq \varepsilon\right) \leq 2d^2 \exp\left(-cm \min\left(\frac{\varepsilon^2}{32^2}, \frac{\varepsilon}{32}\right)\right). \quad (4.26)$$

In particular, for fixed K ,

$$\|B_m^{(K)} - \Lambda_K\|_{\max} = O_{\mathbb{P}}\left(\sqrt{\frac{\log K}{m}}\right). \quad (4.27)$$

Theorem 4.7 shows that, on a fixed truncated Fourier subspace, the frozen finite-width tangent block is a random perturbation of the continuum Fourier block. The perturbation decreases with width at the usual $m^{-1/2}$ concentration scale, up to logarithmic factors in the block dimension.

We next check this concentration behavior numerically. For each width m , we sample 100 independent initializations and compute

$$\|B_m^{(K)} - \Lambda_K\|_{\max}.$$

Figure 4.11 shows the median and 95% quantile of this error, together with the rate envelope suggested by Theorem 4.7. For visualization, we plot the envelope with $c = 0.2$. This is a conservative representative value for the universal constant in the sub-exponential Bernstein inequality; we discuss this choice in Section 5.5.1. The empirical errors decrease steadily with width. The theoretical envelope is much larger than the observed errors, but it has the correct qualitative trend: the finite-width block concentrates around the continuum Fourier block as m increases.

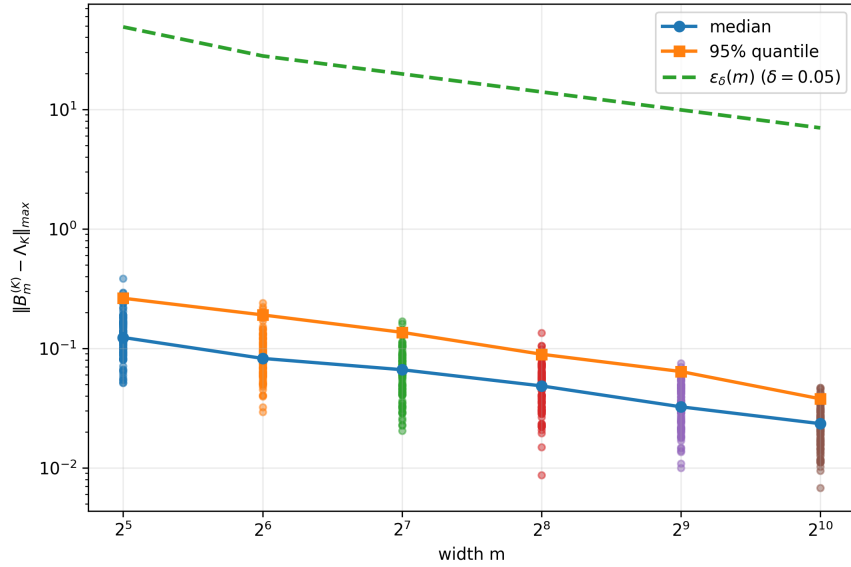


Figure 4.11: Finite-width concentration of the frozen tangent block. For each width m , the plot shows the median and 95% quantile of $\|B_m^{(K)} - \Lambda_K\|_{\max}$ across 100 independent initializations. The dashed curve shows the rate envelope from Theorem 4.7 with $\delta = 0.05$ and the illustrative constant $c = 0.2$. The empirical errors decrease with width, while the explicit envelope is conservative.

4.4.1 Empirical alignment of Fourier frequency subspaces

The block concentration result controls the matrix $B_m^{(K)}$ on the retained Fourier coordinates. We next examine the corresponding eigenspaces directly. Using the quantities introduced in Section 3.5.5, we compare each Fourier frequency subspace $\mathcal{F}_k$ with a matched empirical eigenspace $\widehat{\mathcal{F}}_k$ of the frozen finite-width operator.

We first summarize the comparison using the projector error

$$\varepsilon_{k,F}^{(m)} = \|\widehat{P}_k - P_k\|_F.$$

This gives one scalar mismatch measure for each frequency subspace. By (3.48), it is equivalent to combining the principal angles between $\mathcal{F}_k$ and $\widehat{\mathcal{F}}_k$.

Figure 4.12 shows the mean and standard deviation of $\|\widehat{P}_k - P_k\|_F$ across random initializations. For all displayed frequencies, the projector error decreases as the width increases. Higher frequencies generally have larger projector error at a fixed width. One exception in this experiment is $k = 2$, which has the smallest projector error among the displayed frequencies. We return to this point in the discussion.

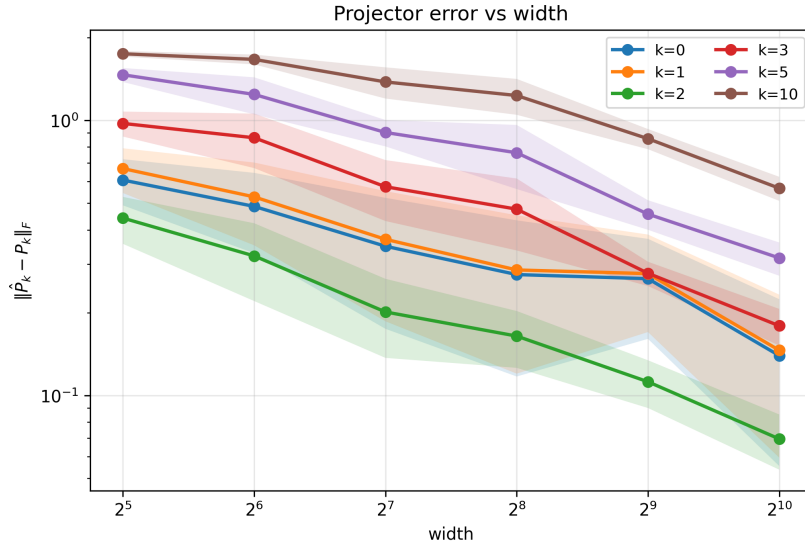


Figure 4.12: Projector error $\|\widehat{P}_k - P_k\|_F$ between the Fourier frequency subspace $\mathcal{F}_k$ and the matched empirical eigenspace $\widehat{\mathcal{F}}_k$ of the frozen finite-width operator at initialization. The markers show the mean over random initializations and the shaded regions show one standard deviation. The error decreases with width for all displayed frequencies.

The projector error gives a single number for each frequency subspace. For $k \geq 1$, however, each frequency subspace is two-dimensional. We therefore also inspect the two principal angles $\theta_{k,1}$ and $\theta_{k,2}$ between $\mathcal{F}_k$ and $\widehat{\mathcal{F}}_k$.

Figure 4.13 shows the mean and standard deviation of these angles across random initializations. The principal angles tend to decrease as the width increases. The lower-frequency subspaces are already relatively well aligned at small width. Higher-frequency

subspaces start with larger angles and improve visibly with m , but they remain less aligned than the lower-frequency subspaces at the same width. As in the projector-error plot, the frequency $k = 2$ is unusually well aligned compared with neighbouring displayed frequencies.

For $k \geq 1$, the second principal angle is usually larger than the first. It therefore gives the stricter test of whether the full two-dimensional Fourier plane has been recovered.

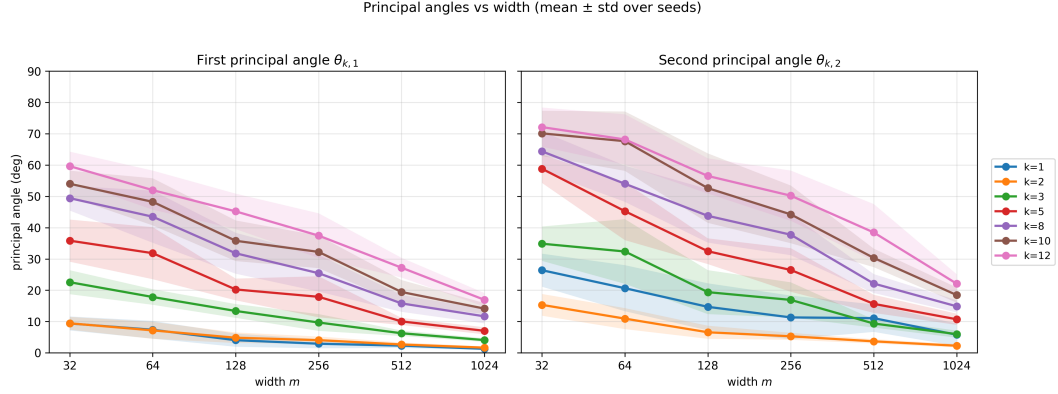


Figure 4.13: Principal angles between the Fourier frequency subspace $\mathcal{F}_k$ and the matched empirical eigenspace $\hat{\mathcal{F}}_k$ of the frozen finite-width operator at initialization. The left panel shows the first principal angle $\theta_{k,1}$, and the right panel shows the second principal angle $\theta_{k,2}$. The markers show the mean over random initializations and the shaded regions show one standard deviation. The principal angles tend to decrease with width.

Taken together, Figures 4.11, 4.12, and 4.13 show the same finite-width trend from complementary viewpoints. The matrix block $B_m^{(K)}$ approaches the continuum block Λ_K as m increases, and the matched empirical eigenspaces become closer to the corresponding Fourier frequency subspaces.

Chapter 5

Discussion

5.1 Overall answer to the main research question

The main research question asked whether the Fourier decomposition of the continuum infinite-width NTK on S^1 persists under finite-sample and finite-width perturbations.

In the continuum infinite-width setting, the Fourier decomposition is exact. Proposition 4.1 shows that the limiting NTK is rotation-invariant on S^1 , so the associated operator is a convolution operator. The Fourier modes therefore diagonalize it, and Theorem 4.2 gives the corresponding eigenvalues. This is the standard NTK spectral mechanism in an explicit setting: components aligned with larger kernel eigenvalues decay faster (Jacot et al., 2020; Lee et al., 2020; Cao et al., 2020). Because the domain and measure are rotationally symmetric, this eigenvalue ordering is also a frequency ordering (Bietti and Mairal, 2019; Ronen et al., 2019; Basri et al., 2020).

The finite-sample and finite-width results show that this Fourier picture is stable only in a qualified sense. After finite sampling, exact diagonalization is lost, but the sampled operator remains close to the diagonal Fourier prediction on fixed low-frequency blocks. At finite width, the frozen tangent operator at initialization is a random perturbation of the continuum Fourier block and concentrates around it as width increases. These results give a controlled perturbation theory around the ideal NTK model.

The part not settled here is the evolving finite-width case. During training, the tangent operator can change, so the frozen Fourier block structure need not remain fixed. The finite-width result should therefore be read as a frozen-kernel baseline, not as a complete theory of finite-width training (Bordelon and Pehlevan, 2023; Vyas et al., 2022; Golikov et al., 2022).

5.2 Q1: finite-sample changes

Q1 asked how replacing the uniform measure on S^1 by n sampled points changes the action of the infinite-width NTK on a fixed low-frequency Fourier subspace, and what this implies for residual dynamics.

5.2.1 From exact to approximate Fourier structure

In the continuum setting, the Fourier basis diagonalizes the NTK operator exactly. After sampling, this is no longer true: the sampled Fourier functions are not exactly orthonormal on the sampled points, and they are not exact eigenvectors of the sampled operator. The finite-sample results quantify these two changes through $G - I_d$, $H - \Lambda^{(n)}$, and $A\Phi - \Phi\Lambda^{(n)}$.

Theorem 4.5 and Figure 4.3 show that these errors decrease with n on a fixed Fourier block. Thus the randomly sampled problem lies between two idealized population settings: the uniform population operator, where harmonic modes diagonalize the kernel, and non-uniform population operators, where the eigendirections become distribution-dependent (Basri et al., 2020; Bietti and Mairal, 2019; Cao et al., 2020). The contribution here is an explicit finite-sample perturbation statement for a prescribed Fourier block.

5.2.2 Implications for residual dynamics

The sampled dynamics in Figures 4.4 and 4.5 show how this approximate structure affects training. Early in training, the residual mode curves follow the finite-sample diagonal prediction. This indicates that the main rate ordering is still governed by the corrected eigenvalues $\lambda_p^{(n)}$.

The later deviations are explained by the decomposition

$$A\Phi = \Phi\Lambda^{(n)} + (A\Phi - \Phi\Lambda^{(n)}).$$

The second term is the part of the sampled operator action that is not explained by the diagonal Fourier model. Early in training, the diagonal decay dominates because the residual components are still large. Later, after some components have become very small, even a small non-diagonal contribution can be visible in the normalized mode curves. This explains why the diagonal prediction captures the initial decay rates better than the late-time behavior.

5.2.3 Preconditioning as a diagnostic

The preconditioning experiment gives a second test of the same interpretation. The theory preconditioner uses only the diagonal prediction $\Lambda^{(n)}$, while the empirical preconditioner uses the observed block $C_n^{(K)} = G^{-1}H$. If both behave similarly, then most of the observed rate separation is explained by eigenvalue scaling. If the empirical preconditioner is substantially better, then finite-sample mixing inside the retained block is important.

Figures 4.7, 4.9, and 4.10 show that the theory preconditioner already removes most of the rate separation on the retained block. This is notable because P_{th} only rescales the retained Fourier coefficients by the predicted eigenvalues. It does not use the observed matrix $C_n^{(K)}$, so it does not correct the finite-sample rotation or mixing of the retained Fourier directions.

The empirical preconditioner is closer to the ideal flattened dynamics because it uses $C_n^{(K)}$. It therefore corrects both the eigenvalue scaling and the mixing caused by the particular sampled points. The fact that P_{th} remains close to P_{emp} in this experiment suggests that

most of the rate separation is explained by the predicted Fourier eigenvalues. The remaining difference is due to the sampled operator not being exactly diagonal in the retained Fourier basis.

This also gives a useful interpretation of the preconditioning experiment. If the target is known to be supported mainly on a fixed range of low frequencies, and if the sampled Fourier block is already close to diagonal, then rescaling by the predicted eigenvalues can make those retained modes converge on more similar time scales.

5.2.4 Answer to Q1

Finite sampling does not preserve the exact Fourier decomposition, but it does preserve it approximately on fixed low-frequency blocks. As n increases, the sampled Fourier basis becomes closer to orthonormal, and applying A to the retained Fourier modes becomes closer to multiplying each mode by its predicted eigenvalue. This is why the corrected eigenvalues predict the main early-time decay rates. The remaining late-time deviations come from the small part of $A\Phi$ that is not captured by the diagonal prediction $\Phi\Lambda^{(n)}$.

5.3 Role of the trainable bias

Corollary 4.3 shows that the trainable hidden bias changes the nullspace of the continuum operator. Without the bias contribution, the odd modes $k \geq 3$ have zero eigenvalue. With the bias contribution, these modes have nonzero but small eigenvalues. Thus the bias determines whether these modes can be learned at all in the fixed-kernel continuum dynamics.

This is consistent with earlier harmonic analyses of two-layer ReLU networks on the sphere and circle, where bias-free two-layer models do not express odd harmonics beyond the first frequency (Ronen et al., 2019; Basri et al., 2020). In the present calculation, the same effect appears directly in the NTK decomposition: removing the trainable hidden-bias contribution sets the odd eigenvalues $k \geq 3$ to zero, while adding it makes them nonzero. Thus Figure 4.2 distinguishes modes that are present with small eigenvalues from modes that are absent because the corresponding eigenvalues vanish.

5.4 Q2: finite-width changes

Q2 asked how finite width changes the continuum operator and the residual dynamics when the input measure remains uniform on S^1 .

5.4.1 Frozen tangent operator as the current baseline

The current finite-width result addresses the frozen tangent operator at initialization. Theorem 4.7 shows that

$$\mathbb{E}[B_m^{(K)}] = \Lambda_K$$

and that $B_m^{(K)}$ concentrates around Λ_K as m increases. Thus, on a fixed low-frequency block, finite width acts as a random perturbation of the continuum Fourier operator.

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Figure 4.11 supports this conclusion. The median and 95% quantile of $\|B_m^{(K)} - \Lambda_K\|_{\max}$ both decrease as m increases. The theoretical envelope is conservative, but the qualitative trend matches the concentration statement. This agrees with the role of the infinite-width NTK as a useful baseline, while also showing why finite-width fluctuations can matter quantitatively (Jacot et al., 2020; Lee et al., 2020; Bordelon and Pehlevan, 2023; Vyas et al., 2022).

5.4.2 Frequency ordering in the alignment diagnostics

The subspace-alignment plots give an eigenspace-level view of the same finite-width effect. Figures 4.12 and 4.13 show that the matched empirical eigenspaces become more aligned with the corresponding Fourier subspaces as m increases. This matches the trend predicted by the concentration result for $B_m^{(K)} - \Lambda_K$.

The ordering across frequencies is not perfectly monotone. In particular, the $k = 2$ Fourier plane is better aligned than some lower-frequency blocks in the projector-error plot. One possible reason is that $k = 0$ and $k = 1$ are structurally special in this model. The constant mode is one-dimensional and receives a direct contribution from the trainable-bias part of the NTK. The first frequency is tied to the coordinate functions in the embedding $x(\varphi) = (\cos \varphi, \sin \varphi)$, and it is the only odd frequency already present in the no-bias kernel:

$$\lambda_1^{(\text{nb})} = \frac{1}{4}, \quad \lambda_k^{(\text{nb})} = 0 \quad \text{for odd } k \geq 3.$$

Thus $k = 0$ and $k = 1$ are not generic representatives of the higher-frequency pattern.

The main conclusion from the alignment diagnostics is the width trend. For each displayed frequency, the matched empirical subspace becomes more aligned with the corresponding Fourier subspace as m increases.

5.4.3 High-frequency sensitivity

The finite-width error should not only be judged by its absolute size. It also matters how large this error is compared with the eigenvalues of the Fourier modes being studied. In the continuum fixed-kernel model, the residual in mode k decays at rate λ_k . For low frequencies, λ_k is relatively large, so a small finite-width perturbation is less likely to change the observed decay. For high frequencies, λ_k is small, so the same perturbation can be large compared with the decay rate of that mode. From Theorem 4.2,

$$\lambda_k = O(k^{-2}) \quad (k \rightarrow \infty),$$

for both even and odd frequencies.

The same issue appears at the eigenspace level. For each $k \geq 1$, the cosine and sine modes form a two-dimensional Fourier plane. At finite width, the matched empirical eigenspace can rotate away from this plane. This rotation is more visible when neighbouring Fourier eigenvalues are close together, which is more common at higher frequencies. Thus high-frequency components are not only slower to learn in the continuum model; they are also

more sensitive to finite sampling and finite-width perturbations. This is consistent with empirical observations that high-frequency components are learned later and are more sensitive to the data distribution and to perturbations (Rahaman et al., 2019; Ronen et al., 2019; Basri et al., 2020).

5.4.4 Answer to Q2

For Q2, the thesis establishes the frozen-kernel part of the answer. For the frozen tangent operator, finite width preserves the continuum Fourier block up to a random perturbation that decreases with m . The alignment diagnostics show the same trend at the level of eigenspaces: the matched empirical subspaces become closer to the Fourier frequency subspaces as width increases. The full answer requires a comparison between frozen and evolving finite-width dynamics.

5.5 Limitations

5.5.1 Scope of the concentration results

The concentration results should be read as fixed-block perturbation results. They control a prescribed low-frequency space $\mathcal{H}_K$ with fixed K . They do not give a uniform statement over all Fourier modes, and the bounds are not designed to remain sharp as K grows with n or m . This is important because the high-frequency eigenvalues and eigengaps become small, so controlling higher and higher frequencies would require increasingly accurate operator approximations.

The numerical constants are also conservative. In the finite-sample theorem, the proof controls entries of the relevant matrices and then applies union bounds over the retained block. The bound for $H - \Lambda^{(n)}$ uses a bounded-differences argument, and the bound for $A\Phi - \Phi\Lambda^{(n)}$ uses a conditional Bernstein estimate followed by a union bound over sample-mode pairs. In the finite-width theorem, each projected entry is controlled using a coarse tail envelope before another union bound is applied. These steps are sufficient to prove concentration, but they discard matrix structure and variance information. This explains why the theoretical envelopes in Figures 4.3 and 4.11 are much larger than the observed errors.

In the finite-width concentration plot, the theorem contains an unspecified universal constant $c > 0$. This constant comes from Bernstein’s inequality for averages of centered sub-exponential random variables (Vershynin, 2018a). In explicit proofs of such inequalities, constants of order $1/4$ often appear after converting a ψ_1 -norm bound into a moment generating function bound. The plotted choice $c = 0.2$ is a conservative representative value, not an optimized constant for this problem. Changing this constant shifts the envelope but does not change the predicted dependence on m .

A sharper analysis would need to use the structure of the objects being controlled. For finite sampling, H is a bounded second-order statistic, so concentration tools for U- or V-statistics may give tighter estimates. For both finite sampling and finite width, direct matrix

concentration in operator norm would be preferable to entrywise control followed by norm conversion.

5.5.2 Limits of the dynamics claims

The finite-sample dynamics experiments support the diagonal prediction, but the theory does not give a complete time-uniform residual bound. The experiments show that the predicted eigenvalues explain the dominant early-time behavior. At later times, some residual components have already become very small, so even a small difference between $A\Phi$ and $\Phi\Lambda^{(n)}$ can become visible in relative terms. The current bounds identify the source of this effect, but they do not give a sharp description of the full residual trajectory for all times.

The preconditioning experiment is also a diagnostic rather than a proposed training method. The theory preconditioner tests whether the observed rate separation is mainly explained by the diagonal eigenvalue scaling. The empirical preconditioner tests how much is gained by using the actual sampled low-frequency block. These experiments support the interpretation of the sampled dynamics, but they do not establish a general preconditioning algorithm or a generalization guarantee.

5.5.3 Limits of the finite-width alignment diagnostics

The projector-error and principal-angle plots are empirical diagnostics of the frozen finite-width operator. They compare Fourier frequency subspaces with matched empirical eigenspaces of a discretized operator. This is useful for checking whether the eigenspaces become more Fourier-like as width increases, but it is not a separate theorem.

The matching procedure also introduces a practical limitation. For each frequency k , the empirical eigenspace is chosen by overlap with $\mathcal{F}_k$. This makes the comparison interpretable, but it also means that the diagnostic depends on the selected discretization grid and on the eigenspace-matching rule. The most useful conclusion from these plots is the trend of errors in frequencies as width increases.

5.5.4 Frozen versus evolving finite-width kernels

The finite-width theorem concerns the tangent operator at initialization. It does not control how the tangent operator changes during training. Therefore it separates one source of finite-width error, namely the random initialization of the tangent kernel, but not the second source, namely kernel evolution during training. A full answer to Q2 would require a time-dependent comparison between the frozen block $B_m^{(K)}$, the evolving block $C_t^{(K)}$, and the continuum block Λ_K .

There is also no joint theorem combining finite sampling, finite width, and training time. The finite-sample analysis studies the infinite-width sampled operator. The finite-width analysis studies the continuum projected frozen operator. In real-life experiments, these effects occur together. A complete perturbation theory would need to control both at the same time.

5.5.5 Model assumptions

The analysis relies on the special structure of S^1 with the uniform measure. In this setting, the limiting kernel is rotation-invariant, so the corresponding operator is a convolution operator and the Fourier basis diagonalizes it exactly. Related harmonic descriptions of neural tangent kernels on the circle or sphere appear in Ronen et al. (2019); Basri et al. (2020); Bietti and Mairal (2019); Cao et al. (2020). This structure is no longer automatic once the input distribution is changed. For example, under a non-uniform density on the circle, the integral operator is weighted by the density and is no longer diagonal in the global Fourier basis; the relevant eigenfunctions become distribution-dependent, as studied by Basri et al. (2020). If the input domain is changed, the appropriate basis also changes and may not be available in closed form.

The architecture and scaling assumptions are also part of the result. The thesis uses a two-layer ReLU network in NTK scaling. The ReLU activation is important because its Gaussian averages have explicit arc-cosine formulas (Cho and Saul, 2009), which makes the kernel and its Fourier coefficients computable in closed form. The $1/\sqrt{m}$ scaling is important because it places the model in the NTK or lazy-training regime, where the infinite-width limit is described by a fixed kernel (Jacot et al., 2020; Lee et al., 2020; Golikov et al., 2022). Changing the activation, adding or removing bias terms, changing the parameterization, or moving to a feature-learning scaling changes the limiting kernel and therefore changes the spectrum that drives the dynamics. Such feature-learning regimes are better described by mean-field or related infinite-width limits (Mei et al., 2018; Chizat and Bach, 2018; Sirignano and Spiliopoulos, 2019; Rotskoff and Vanden-Eijnden, 2022). Thus the Fourier conclusions should be understood as conclusions about this explicit NTK model, not as architecture-independent statements about neural networks in general.

Chapter 6

Conclusion and future work

6.1 Conclusion

This thesis studied spectral bias in a controlled NTK setting: a two-layer ReLU network on the unit circle S^1 . In the continuum infinite-width limit with the uniform measure, the limiting NTK is rotation-invariant. The corresponding operator is therefore a convolution operator, and the Fourier basis diagonalizes it exactly. This gives an explicit fixed-kernel description of training: each Fourier component of the residual decays independently at a rate determined by the corresponding kernel eigenvalue.

The first part of the thesis studied what changes after replacing the continuum measure by finitely many sampled points. Exact Fourier diagonalization is lost, but the Fourier description remains accurate on fixed low-frequency blocks. The sampled Fourier Gram matrix G , the matrix H describing the sampled operator on the retained Fourier coordinates, and the difference $A\Phi - \Phi\Lambda^{(n)}$ all become smaller as the sample size increases. This explains why the finite-sample corrected eigenvalues predict the dominant early-time residual dynamics. The preconditioning experiment supports the same interpretation: using only the diagonal eigenvalue prediction removes most of the observed rate separation on the retained block.

The second part of the thesis studied finite width in the frozen-kernel setting. For each fixed Fourier block, the frozen finite-width tangent operator at initialization has expectation equal to the continuum Fourier block and concentrates around it as the width increases. The numerical experiments show the same trend both at the matrix level and at the eigenspace level: the matched empirical eigenspaces become more aligned with the corresponding Fourier frequency subspaces as width increases.

The main conclusion is that the Fourier picture is robust, but only in a qualified sense. It is exact for the continuum infinite-width operator. It remains accurate after finite sampling on fixed low-frequency blocks. It also remains accurate for the frozen finite-width tangent operator when the width is large. The thesis does not prove the corresponding statement for the evolving finite-width tangent kernel during training.

A recurring theme is that high-frequency modes are more sensitive to perturbations. In the continuum model, high frequencies are learned slowly because their eigenvalues

are small. Under finite sampling or finite width, the same small eigenvalues make high-frequency components more affected by small changes in the operator. Thus, in this model, high frequencies are not only slower to learn; their Fourier structure is also less stable under sampling and finite-width perturbations.

6.2 Future work

The most direct next step is to complete the finite-width part of the thesis by comparing frozen and evolving tangent-kernel dynamics. This would separate the effect of random initialization of the finite-width kernel from the effect of kernel evolution during training.

A second direction is to prove a joint perturbation result that includes finite sampling, finite width, and training time. The present thesis studies the infinite-width sampled operator and the continuum frozen finite-width operator separately. In real-life experiments, these effects occur together.

A third direction is to sharpen the concentration bounds. The current proofs use entry-wise concentration, coarse tail bounds, and union bounds. These arguments are sufficient to prove the correct decreasing trends, but they are conservative as numerical estimates. Direct operator-norm bounds, variance-sensitive estimates, or concentration inequalities for U- and V-statistics may give bounds closer to the observed errors.

A fourth direction is to study the alignment diagnostics more directly. The projector-error and principal-angle plots suggest that different Fourier subspaces can have different finite-width stability. A more detailed analysis could measure which Fourier blocks are most strongly mixed by the finite-width operator and how this depends on the eigenvalues and gaps of the continuum operator.

Finally, the analysis should be extended beyond the uniform circle setting. Non-uniform input densities, higher-dimensional spheres, other activations, and different parameterizations all change the relevant kernel operator and its spectrum. Studying these cases would clarify which conclusions are special to the explicit Fourier model on S^1 , and which reflect a broader mechanism behind spectral bias.

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Appendix A

Mean-Field Viewpoint

A.1 Detailed derivation of the mean-field representation

A.1.1 Parameter space and empirical measure

Let $\Omega := \mathbb{R} \times \mathbb{R}^d$ denote the parameter space, and let $\mathcal{F}$ be its Borel σ -algebra. For each neuron α , define the Dirac measure δ_{θ_α} on $(\Omega, \mathcal{F})$, i.e. if $A \in \mathcal{F}$ then,

$$\delta_{\theta_\alpha}(A) = \begin{cases} 1, & \theta_\alpha \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Define the empirical neuronal measure

$$\mu_m := \frac{1}{m} \sum_{\alpha=1}^m \delta_{\theta_\alpha}. \quad (\text{A.1})$$

By construction, $\mu_m \geq 0$ and $\mu_m(\Omega) = 1$, hence $\mu_m \in \mathcal{P}(\Omega)$ where $\mathcal{P}(\Omega)$ is the space of Borel probability measures on Ω .

Integration against Dirac measures. Let χ_A denote the indicator function of a measurable set $A \subset \Omega$. Then, for any $\theta_\alpha \in \Omega$,

$$\int_{\Omega} \chi_A(\theta) d\delta_{\theta_\alpha}(\theta) = \delta_{\theta_\alpha}(A) = \chi_A(\theta_\alpha).$$

More generally, if $s(\theta) = \sum_{k=1}^r c_k \chi_{A_k}(\theta)$ is a simple function, then

$$\int_{\Omega} s(\theta) d\delta_{\theta_\alpha}(\theta) = \sum_{k=1}^r c_k \chi_{A_k}(\theta_\alpha) = s(\theta_\alpha).$$

Any nonnegative measurable function $g : \Omega \rightarrow \mathbb{R}$ can be approximated by a sequence of simple functions, $\{s_n\}_{n \geq 0}$. If $s_n \uparrow g$ and each $s_n \geq 0$ then by the Monotone Convergence Theorem,

$$\int_{\Omega} g d\delta_{\theta_\alpha} = \int_{\Omega} \lim_{n \rightarrow \infty} s_n d\delta_{\theta_\alpha} = \lim_{n \rightarrow \infty} \int_{\Omega} s_n d\delta_{\theta_\alpha} = \lim_{n \rightarrow \infty} s_n(\theta_\alpha) = g(\theta_\alpha). \quad (\text{A.2})$$

Integral representation of the network. Consider now the integral of g with respect to the empirical measure μ_m :

$$\int_{\Omega} g d\mu_m = \int_{\Omega} g d\left(\frac{1}{m} \sum_{\alpha=1}^m \delta_{\theta_{\alpha}}\right) = \frac{1}{m} \sum_{\alpha=1}^m g(\theta_{\alpha}).$$

Define

$$g(\theta_{\alpha}) = g(v_{\alpha}, w_{\alpha}) := v_{\alpha} \Phi(w_{\alpha}^{\top} x).$$

Then

$$\int_{\Omega} g d\mu_m = \frac{1}{m} \sum_{\alpha=1}^m v_{\alpha} \Phi(w_{\alpha}^{\top} x) = f_{\theta}(x), \quad (\text{A.3})$$

recovering the finite-width network.

A.1.2 Initialization and weak convergence

Assume that the parameters $\{\theta_{\alpha}(0)\}_{\alpha=1}^m$ are initialized i.i.d. according to a probability measure μ_0 on Ω . Define the empirical measure at initialization

$$\mu_{m,0} := \frac{1}{m} \sum_{\alpha=1}^m \delta_{\theta_{\alpha}(0)}.$$

Then $\mu_{m,0}$ converges weakly to μ_0 almost surely as $m \rightarrow \infty$. Indeed, for any bounded continuous test function $\psi : \Omega \rightarrow \mathbb{R}$,

$$\int_{\Omega} \psi d\mu_{m,0} = \frac{1}{m} \sum_{\alpha=1}^m \psi(\theta_{\alpha}(0)) \xrightarrow[m \rightarrow \infty]{\text{a.s.}} \mathbb{E}_{\theta \sim \mu_0}[\psi(\theta)] = \int_{\Omega} \psi d\mu_0,$$

where the convergence follows from the strong law of large numbers applied to the i.i.d. random variables $\psi(\theta_{\alpha}(0))$. Since this holds for all bounded continuous ψ , we conclude that $\mu_{m,0} \rightarrow \mu_0$ (Varadarajan, 1958). As a consequence, the network output at initialization converges pointwise to the deterministic limit

$$f_{\mu_0}(x) := \int_{\Omega} v \Phi(w^{\top} x) d\mu_0(v, w).$$

For instance, under i.i.d. Gaussian initialization of the parameters, i.e. $v_0 \sim \mathcal{N}(0, I_m)$ $W_0 \sim \mathcal{N}(0, I_m \otimes I_d)$, the limiting initial measure $\mu_0 = \mathcal{N}(0, I_{d+1})$ is a Gaussian measure on a $d+1$ -dimensional space.

A.1.3 Training dynamics and evolution of the empirical measure

Let $X = (x_1, \dots, x_n) \in \mathbb{R}^{d \times n}$ and $y = (y_1, \dots, y_n) \in \mathbb{R}^n$ denote the training set, and consider the squared loss

$$L(v, W) = \frac{1}{2} \sum_{i=1}^n (f_{v,W}(x_i) - y_i)^2 = \frac{1}{2} \|f_{v,W}(X) - y\|_2^2,$$

where $f_{v,W}(X) := (f_{v,W}(x_i))_{i=1}^n$.

We study continuous-time gradient flow for a finite-width network with m neurons and learning rate $\eta > 0$,

$$\dot{v}_{t,\alpha} = -\eta \nabla_{v_\alpha} L(v_t, W_t), \quad \dot{w}_{t,\alpha} = -\eta \nabla_{w_\alpha} L(v_t, W_t), \quad \alpha \in \{1, \dots, m\}.$$

A direct computation yields, for each α ,

$$\dot{v}_{t,\alpha} = \frac{\eta}{m} \varphi(X^\top w_{t,\alpha})^\top (y - f_{v_t, W_t}(X)), \quad (\text{A.4})$$

$$\dot{w}_{t,\alpha} = \frac{\eta}{m} X \left(\varphi'(X^\top w_{t,\alpha}) \odot (y - f_{v_t, W_t}(X)) \right) v_{t,\alpha}, \quad (\text{A.5})$$

where $\odot$ denotes elementwise (Hadamard) multiplication.

As seen in (A.4)–(A.5), evolution of each neuron depends on the parameters only through the current empirical measure $\mu_{m,t}$ via $f_{v_t, W_t}(X)$. Thus, the particle system (A.4)–(A.5) induces an evolution of the empirical measure in parameter space (Mei et al., 2018; Rotskoff and Vanden-Eijnden, 2022).

Hence, the particle system (A.4)–(A.5) can be viewed as an interacting particle system driven by a measure-dependent velocity field. For any $\theta = (v, w) \in \Omega$ and any $\mu \in \mathcal{P}(\Omega)$, define

$$b((v, w); \mu) := \begin{pmatrix} \varphi(X^\top w)^\top (y - f_\mu(X)) \\ X \left(\varphi'(X^\top w) \odot (y - f_\mu(X)) \right) v \end{pmatrix},$$

where

$$f_\mu(x) := \int_{\Omega} v \varphi(w^\top x) d\mu(v, w), \quad f_\mu(X) := (f_\mu(x_i))_{i=1}^n.$$

Then the gradient-flow dynamics can be written compactly as

$$\dot{\theta}_{t,\alpha} = \frac{\eta}{m} b(\theta_{t,\alpha}; \mu_{m,t}), \quad \alpha = 1, \dots, m.$$

In particular, each particle interacts with the rest of the system only through the current empirical measure $\mu_{m,t}$ (via $f_{\mu_{m,t}}(X)$).

A.1.4 Continuity equation and push-forward formulation

Equivalently, $\mu_{m,t}$ solves the continuity equation

$$\partial_t \mu_{m,t} + \nabla_\theta \cdot \left(\mu_{m,t} \frac{\eta}{m} b(\cdot; \mu_{m,t}) \right) = 0, \quad (\text{A.6})$$

in the sense of distributions, i.e. for every test function $\psi \in C_c^\infty(\Omega)$,

$$\frac{d}{dt} \int_{\Omega} \psi(\theta) d\mu_{m,t}(\theta) = \frac{\eta}{m} \int_{\Omega} \nabla_\theta \psi(\theta) \cdot b(\theta; \mu_{m,t}) d\mu_{m,t}(\theta).$$

Choosing the mean-field scaling $\eta = m$ yields a nontrivial $O(1)$ evolution in time and, as $m \rightarrow \infty$, one expects $\mu_{m,t} \rightarrow \mu_t$, where the limit μ_t satisfies

$$\partial_t \mu_t + \nabla_\theta \cdot (\mu_t b(\cdot; \mu_t)) = 0, \quad \mu_0 = \mathcal{N}(0, I_{d+1}). \quad (\text{A.7})$$

Push-forward formulation. Recall that if $g : \mathcal{X} \rightarrow \mathcal{Z}$ is measurable and $\mu \in \mathcal{P}(\mathcal{X})$, then the *push-forward* of μ by g is the measure $g_*\mu \in \mathcal{P}(\mathcal{Z})$ defined by

$$(g_*\mu)(A) := \mu(g^{-1}(A)), \quad A \subset \mathcal{Z} \text{ measurable.}$$

Let $\Phi_t : \Omega \rightarrow \Omega$ denote the flow map associated with the velocity field $\frac{\eta}{m}b(\cdot; \mu_{m,t})$. Then the empirical measure is transported by this flow:

$$\mu_{m,t} = (\Phi_t)_*\mu_{m,0}.$$

In particular, for any measurable observable $h : \Omega \rightarrow \mathbb{R}$,

$$\int_{\Omega} h(\theta) d\mu_{m,t}(\theta) = \int_{\Omega} h(\Phi_t(\theta)) d\mu_{m,0}(\theta),$$

so evaluating the network at time t is obtained by taking $h_x(\theta) := v\varphi(w^\top x)$ and writing

$$f_{\mu_{m,t}}(x) = \int_{\Omega} v\varphi(w^\top x) d\mu_{m,t}(v, w) = \int_{\Omega} v\varphi(w^\top x) d((\Phi_t)_*\mu_{m,0})(v, w).$$

Appendix B

Derivation of the limiting kernel on S^1

This appendix derives the explicit form of the limiting neural tangent kernel on the circle for the model introduced in Chapter 3, and computes the corresponding Fourier eigenvalues of the integral operator T . These derivations justify Proposition 4.1 and Theorem 4.2.

B.1 Decomposition of the empirical neural tangent kernel

Recall the network

$$f(x; \theta) = \frac{1}{\sqrt{m}} \sum_{i=1}^m a_i \sigma(w_i^\top x + b_i),$$

where $x \in S^1 \subset \mathbb{R}^2$, $\sigma(t) = t_+$, and $\theta = \{(a_i, w_i, b_i)\}_{i=1}^m$. For $x, y \in S^1$, the empirical neural tangent kernel is

$$\Theta_m(x, y) = \langle \nabla_\theta f(x; \theta), \nabla_\theta f(y; \theta) \rangle. \quad (\text{B.1})$$

For each hidden unit i , define

$$u_i := w_i^\top x + b_i, \quad v_i := w_i^\top y + b_i.$$

Then

$$\frac{\partial f(x; \theta)}{\partial a_i} = \frac{1}{\sqrt{m}} \sigma(u_i), \quad \frac{\partial f(x; \theta)}{\partial w_i} = \frac{1}{\sqrt{m}} a_i \sigma'(u_i) x, \quad \frac{\partial f(x; \theta)}{\partial b_i} = \frac{1}{\sqrt{m}} a_i \sigma'(u_i).$$

Substituting into (B.1) yields

$$\Theta_m(x, y) = \Theta_m^{(a)}(x, y) + \Theta_m^{(w)}(x, y) + \Theta_m^{(b)}(x, y), \quad (\text{B.2})$$

where

$$\begin{aligned}\Theta_m^{(a)}(x, y) &= \frac{1}{m} \sum_{i=1}^m \sigma(u_i) \sigma(v_i), \\ \Theta_m^{(w)}(x, y) &= \frac{1}{m} \sum_{i=1}^m a_i^2 \sigma'(u_i) \sigma'(v_i) x^\top y, \\ \Theta_m^{(b)}(x, y) &= \frac{1}{m} \sum_{i=1}^m a_i^2 \sigma'(u_i) \sigma'(v_i).\end{aligned}$$

B.2 Infinite-width limit

At initialization, the parameters satisfy

$$a_i \sim \mathcal{N}(0, 1), \quad w_i \sim \mathcal{N}(0, I_2), \quad b_i(0) = 0,$$

independently across i . Since the hidden units are independent and identically distributed, the law of large numbers implies that $\Theta_m(x, y)$ converges almost surely, as $m \rightarrow \infty$, to a deterministic kernel $\Theta(x, y)$.

Because $b_i(0) = 0$, the limiting kernel is

$$\Theta(x, y) = \mathbb{E}[\sigma(u)\sigma(v)] + (x^\top y + 1)\mathbb{E}[\sigma'(u)\sigma'(v)], \quad (\text{B.3})$$

where

$$u = w^\top x, \quad v = w^\top y, \quad w \sim \mathcal{N}(0, I_2).$$

We write

$$K_0(x, y) := \mathbb{E}[\sigma(u)\sigma(v)], \quad K_1(x, y) := \mathbb{E}[\sigma'(u)\sigma'(v)],$$

so that

$$\Theta(x, y) = K_0(x, y) + (x^\top y + 1)K_1(x, y).$$

B.3 Restriction to the circle

Write

$$x(\varphi) = (\cos \varphi, \sin \varphi), \quad x(\psi) = (\cos \psi, \sin \psi),$$

and let $\delta \in [0, \pi]$ denote the angle between $x(\varphi)$ and $x(\psi)$, so that

$$x(\varphi)^\top x(\psi) = \cos \delta.$$

By rotational invariance of the Gaussian measure, the kernel depends only on δ . We therefore write $\Theta(\delta)$, $K_0(\delta)$, and $K_1(\delta)$.

To compute these quantities, we rotate coordinates so that

$$x = (1, 0), \quad y = (\cos \delta, \sin \delta).$$

If $w = (Z_1, Z_2)$ with $Z_1, Z_2 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, then

$$u = Z_1, \quad v = Z_1 \cos \delta + Z_2 \sin \delta.$$

Lemma B.1. For $\delta \in [0, \pi]$,

$$K_1(\delta) = \mathbb{P}(u > 0, v > 0) = \frac{\pi - \delta}{2\pi}. \quad (\text{B.4})$$

Proof. Since (u, v) is a centered bivariate Gaussian with correlation $\cos \delta$, the probability that both coordinates are positive is the standard Gaussian quadrant probability, which equals $(\pi - \delta)/(2\pi)$. $\square$

Lemma B.2. For $\delta \in [0, \pi]$,

$$K_0(\delta) = \mathbb{E}[\sigma(u)\sigma(v)] = \frac{1}{2\pi} \left(\sin \delta + (\pi - \delta) \cos \delta \right). \quad (\text{B.5})$$

Proof. This is the standard arc-cosine covariance formula for ReLU features in two dimensions. $\square$

Proof of Proposition 4.1. Using (B.3), (B.4), and (B.5), we obtain

$$\begin{aligned} \Theta(\delta) &= \frac{1}{2\pi} \left(\sin \delta + (\pi - \delta) \cos \delta \right) + \frac{1 + \cos \delta}{2\pi} (\pi - \delta) \\ &= \frac{1}{2\pi} \left(\sin \delta + 2(\pi - \delta) \cos \delta + (\pi - \delta) \right). \end{aligned} \quad (\text{B.6})$$

This is exactly (4.1). The convolution form of the operator follows from the fact that Θ depends only on the angular difference. $\square$

B.4 Fourier coefficients of the limiting kernel

Because the operator T is a convolution operator, it is diagonalized by the real Fourier basis. If λ_k denotes the eigenvalue associated with frequency k , then

$$\lambda_k = \frac{1}{\pi} \int_0^\pi \Theta(\delta) \cos(k\delta) d\delta, \quad k \geq 0. \quad (\text{B.7})$$

Substituting (B.6) gives

$$\lambda_k = \frac{1}{2\pi^2} \int_0^\pi \left(\sin \delta + 2(\pi - \delta) \cos \delta + (\pi - \delta) \right) \cos(k\delta) d\delta.$$

We first record a basic integration lemma.

Lemma B.3. For every integer $m \geq 1$,

$$I_m := \int_0^\pi (\pi - \delta) \cos(m\delta) d\delta = \frac{1 - (-1)^m}{m^2}. \quad (\text{B.8})$$

Proof. Integrating by parts gives

$$I_m = \left[(\pi - \delta) \frac{\sin(m\delta)}{m} \right]_0^\pi + \frac{1}{m} \int_0^\pi \sin(m\delta) d\delta.$$

The boundary term vanishes, and

$$\frac{1}{m} \int_0^\pi \sin(m\delta) d\delta = \frac{1}{m} \left[-\frac{\cos(m\delta)}{m} \right]_0^\pi = \frac{1 - (-1)^m}{m^2}.$$

□

Proof of Theorem 4.2. For $k \geq 2$, write

$$\lambda_k = \frac{1}{2\pi^2} (A_k + B_k + C_k),$$

where

$$\begin{aligned} A_k &:= \int_0^\pi \sin \delta \cos(k\delta) d\delta, \\ B_k &:= \int_0^\pi 2(\pi - \delta) \cos \delta \cos(k\delta) d\delta, \\ C_k &:= \int_0^\pi (\pi - \delta) \cos(k\delta) d\delta. \end{aligned}$$

Using

$$\sin \delta \cos(k\delta) = \frac{1}{2} (\sin((k+1)\delta) - \sin((k-1)\delta)),$$

we obtain

$$A_k = \frac{1}{2} \left(\frac{1 - (-1)^{k+1}}{k+1} - \frac{1 - (-1)^{k-1}}{k-1} \right).$$

Using

$$2 \cos \delta \cos(k\delta) = \cos((k+1)\delta) + \cos((k-1)\delta),$$

together with Lemma B.3, we obtain

$$B_k = \frac{1 - (-1)^{k+1}}{(k+1)^2} + \frac{1 - (-1)^{k-1}}{(k-1)^2}, \quad C_k = \frac{1 - (-1)^k}{k^2}.$$

If $k \geq 3$ is odd, then $k \pm 1$ are even, so $A_k = B_k = 0$, while $C_k = 2/k^2$. Hence

$$\lambda_k = \frac{1}{\pi^2 k^2}, \quad k \geq 3 \text{ odd.} \tag{B.9}$$

If $k \geq 2$ is even, then $k \pm 1$ are odd, so $C_k = 0$, and

$$A_k = -\frac{2}{k^2 - 1}, \quad B_k = \frac{4(k^2 + 1)}{(k^2 - 1)^2}.$$

Therefore

$$A_k + B_k = \frac{2(k^2 + 3)}{(k^2 - 1)^2},$$

and so

$$\lambda_k = \frac{k^2 + 3}{\pi^2(k^2 - 1)^2}, \quad k \geq 2 \text{ even.} \quad (\text{B.10})$$

It remains to compute $k = 0$ and $k = 1$ directly. For $k = 0$,

$$\lambda_0 = \frac{1}{2\pi^2} \left(\int_0^\pi \sin \delta d\delta + 2 \int_0^\pi (\pi - \delta) \cos \delta d\delta + \int_0^\pi (\pi - \delta) d\delta \right) = \frac{1}{4} + \frac{3}{\pi^2}.$$

For $k = 1$,

$$\lambda_1 = \frac{1}{2\pi^2} \left(\int_0^\pi \sin \delta \cos \delta d\delta + 2 \int_0^\pi (\pi - \delta) \cos^2 \delta d\delta + \int_0^\pi (\pi - \delta) \cos \delta d\delta \right) = \frac{1}{4} + \frac{1}{\pi^2}.$$

This proves (4.3) and (4.4). $\square$

B.5 Effect of the bias contribution

We now isolate the effect of the trainable hidden bias. From the NTK decomposition in (B.2), the limiting kernel can be written as

$$\Theta(x, y) = K_0(x, y) + x^\top y K_1(x, y) + K_1(x, y).$$

The first two terms come from differentiating with respect to the output weights and input weights. The last term comes from differentiating with respect to the hidden biases. Thus, if the trainable hidden-bias contribution is removed, the limiting kernel becomes

$$\Theta^{(\text{nb})}(x, y) = K_0(x, y) + x^\top y K_1(x, y).$$

On the circle, $x^\top y = \cos \delta$, so by (B.5) and (B.4),

$$\begin{aligned} \Theta^{(\text{nb})}(\delta) &= \frac{1}{2\pi} \left(\sin \delta + (\pi - \delta) \cos \delta \right) + \cos \delta \frac{\pi - \delta}{2\pi} \\ &= \frac{1}{2\pi} \left(\sin \delta + 2(\pi - \delta) \cos \delta \right). \end{aligned} \quad (\text{B.11})$$

The Fourier eigenvalues of this no-bias kernel are

$$\lambda_k^{(\text{nb})} = \frac{1}{\pi} \int_0^\pi \Theta^{(\text{nb})}(\delta) \cos(k\delta) d\delta.$$

Using (B.11), this becomes

$$\lambda_k^{(\text{nb})} = \frac{1}{2\pi^2} \int_0^\pi \left(\sin \delta + 2(\pi - \delta) \cos \delta \right) \cos(k\delta) d\delta.$$

In the notation of the proof of Theorem 4.2,

$$\lambda_k^{(\text{nb})} = \frac{1}{2\pi^2}(A_k + B_k),$$

whereas the full eigenvalue is

$$\lambda_k = \frac{1}{2\pi^2}(A_k + B_k + C_k).$$

Thus the term C_k , which comes from

$$\frac{1}{2\pi}(\pi - \delta),$$

is precisely the trainable-bias contribution.

For $k = 0$, direct integration gives

$$\lambda_0^{(\text{nb})} = \frac{1}{2\pi^2} \left(\int_0^\pi \sin \delta d\delta + 2 \int_0^\pi (\pi - \delta) \cos \delta d\delta \right).$$

Since

$$\int_0^\pi \sin \delta d\delta = 2, \quad \int_0^\pi (\pi - \delta) \cos \delta d\delta = 2,$$

we obtain

$$\lambda_0^{(\text{nb})} = \frac{1}{2\pi^2}(2 + 4) = \frac{3}{\pi^2}.$$

For $k = 1$,

$$\lambda_1^{(\text{nb})} = \frac{1}{2\pi^2} \left(\int_0^\pi \sin \delta \cos \delta d\delta + 2 \int_0^\pi (\pi - \delta) \cos^2 \delta d\delta \right).$$

The first integral is zero. For the second, using

$$\cos^2 \delta = \frac{1 + \cos(2\delta)}{2},$$

we get

$$2 \int_0^\pi (\pi - \delta) \cos^2 \delta d\delta = \int_0^\pi (\pi - \delta) d\delta + \int_0^\pi (\pi - \delta) \cos(2\delta) d\delta.$$

The second term is zero by Lemma B.3, since 2 is even. Hence

$$2 \int_0^\pi (\pi - \delta) \cos^2 \delta d\delta = \frac{\pi^2}{2},$$

and therefore

$$\lambda_1^{(\text{nb})} = \frac{1}{4}.$$

For $k \geq 2$, we use the values of A_k and B_k computed in the proof of Theorem 4.2. If $k \geq 3$ is odd, then $k - 1$ and $k + 1$ are even, so

$$A_k = 0, \quad B_k = 0.$$

Therefore

$$\lambda_k^{(\text{nb})} = 0, \quad k \geq 3 \text{ odd.} \quad (\text{B.12})$$

If $k \geq 2$ is even, then

$$A_k = -\frac{2}{k^2 - 1}, \quad B_k = \frac{4(k^2 + 1)}{(k^2 - 1)^2}.$$

Thus

$$A_k + B_k = \frac{2(k^2 + 3)}{(k^2 - 1)^2},$$

and hence

$$\lambda_k^{(\text{nb})} = \frac{k^2 + 3}{\pi^2(k^2 - 1)^2}, \quad k \geq 2 \text{ even.} \quad (\text{B.13})$$

This proves Corollary 4.3. Comparing the no-bias eigenvalues with the full eigenvalues in Theorem 4.2, the trainable-bias contribution is

$$\lambda_0 - \lambda_0^{(\text{nb})} = \frac{1}{4}, \quad \lambda_1 - \lambda_1^{(\text{nb})} = \frac{1}{\pi^2},$$

while for $k \geq 2$,

$$\lambda_k - \lambda_k^{(\text{nb})} = \begin{cases} 0, & k \geq 2 \text{ even,} \\ \frac{1}{\pi^2 k^2}, & k \geq 3 \text{ odd.} \end{cases}$$

Thus the bias term does not change the even frequencies $k \geq 2$, but it is exactly what makes the odd frequencies $k \geq 3$ visible in the fixed-kernel dynamics. The frequency $k = 1$ is exceptional: it already has eigenvalue $1/4$ without the bias, and the bias adds the additional $1/\pi^2$ contribution.

Appendix C

Proofs of the finite-sample estimates

This appendix proves the finite-sample results from Section 4.2. We work with the truncated Fourier subspace H_K , the feature matrix Φ , the sampled kernel matrix A , the Gram matrix G , and the matrix H defined in (3.19)–(3.22).

C.1 Preliminaries

For each basis function in (3.13)–(3.14),

$$|\phi_p(\varphi)| \leq \sqrt{2} \quad \text{for all } \varphi \in [0, 2\pi), \quad 1 \leq p \leq d. \quad (\text{C.1})$$

We also write

$$\kappa := \sup_{\theta \in [0, 2\pi)} |\Theta(\theta)| < \infty. \quad (\text{C.2})$$

For each basis index p , let λ_p denote the eigenvalue of the operator T associated with ϕ_p , so that

$$T\phi_p = \lambda_p \phi_p.$$

Define the finite-sample corrected eigenvalues by

$$\lambda_p^{(n)} = \left(1 - \frac{1}{n}\right) \lambda_p + \frac{\Theta(0)}{n}, \quad (\text{C.3})$$

and let

$$\Lambda^{(n)} = \text{diag}(\lambda_p^{(n)}).$$

C.2 Expectation of H

Proof of Proposition 4.4. By definition,

$$H_{pq} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \phi_p(\varphi_i) \Theta(\varphi_i - \varphi_j) \phi_q(\varphi_j). \quad (\text{C.4})$$

Split the sum into the cases $i \neq j$ and $i = j$:

$$H_{pq} = \frac{1}{n^2} \sum_{i \neq j} \phi_p(\varphi_i) \Theta(\varphi_i - \varphi_j) \phi_q(\varphi_j) + \frac{1}{n^2} \sum_{i=1}^n \phi_p(\varphi_i) \Theta(0) \phi_q(\varphi_i).$$

For the diagonal terms,

$$\mathbb{E} \left[\frac{1}{n^2} \sum_{i=1}^n \phi_p(\varphi_i) \Theta(0) \phi_q(\varphi_i) \right] = \frac{\Theta(0)}{n} \delta_{pq}.$$

For the off-diagonal terms, independence of φ_i and φ_j gives

$$\begin{aligned} \mathbb{E}[\phi_p(\varphi_i) \Theta(\varphi_i - \varphi_j) \phi_q(\varphi_j)] &= \int_0^{2\pi} \phi_p(\varphi) \left(\int_0^{2\pi} \Theta(\varphi - \psi) \phi_q(\psi) d\mu(\psi) \right) d\mu(\varphi) \\ &= \int_0^{2\pi} \phi_p(\varphi) (T\phi_q)(\varphi) d\mu(\varphi) = \lambda_q \delta_{pq}. \end{aligned}$$

Since there are $n(n-1)$ off-diagonal pairs, we conclude that

$$\mathbb{E}[H_{pq}] = \left(1 - \frac{1}{n}\right) \lambda_q \delta_{pq} + \frac{\Theta(0)}{n} \delta_{pq} = \lambda_q^{(n)} \delta_{pq}.$$

Therefore $\mathbb{E}[H] = \Lambda^{(n)}$. □

C.3 Concentration of the Gram matrix

Proof of (4.13). For fixed p, q ,

$$G_{pq} = \frac{1}{n} \sum_{i=1}^n Z_i, \quad Z_i := \phi_p(\varphi_i) \phi_q(\varphi_i).$$

The variables Z_i are independent and identically distributed, with

$$\mathbb{E}[Z_i] = \delta_{pq}$$

by orthonormality of the Fourier basis.

Set

$$Y_i := Z_i - \delta_{pq}.$$

Then $\mathbb{E}[Y_i] = 0$ and

$$G_{pq} - \delta_{pq} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

By (C.1),

$$|Z_i| \leq 2.$$

If $p \neq q$, then $\delta_{pq} = 0$, so $|Y_i| = |Z_i| \leq 2$. If $p = q$, then

$$Y_i = \phi_p(\varphi_i)^2 - 1,$$

and since $0 \leq \phi_p(\varphi_i)^2 \leq 2$, we have $|Y_i| \leq 1 \leq 2$. Thus

$$|Y_i| \leq 2$$

for all i .

Next,

$$\text{Var}(Y_i) \leq \mathbb{E}[Y_i^2] \leq \mathbb{E}[Z_i^2].$$

Using again (C.1),

$$Z_i^2 = \phi_p(\varphi_i)^2 \phi_q(\varphi_i)^2 \leq 2 \phi_p(\varphi_i)^2.$$

Taking expectations and using orthonormality gives

$$\mathbb{E}[Z_i^2] \leq 2 \int \phi_p(\varphi)^2 d\mu(\varphi) = 2.$$

Hence

$$\text{Var}(Y_i) \leq 2.$$

We now apply Bernstein's inequality Vershynin (2018a) in the following form: if $Y_1, \dots, Y_n$ are independent, centered random variables such that

$$|Y_i| \leq M, \quad \text{Var}(Y_i) \leq \sigma^2$$

for every i , then for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n Y_i\right| \geq \varepsilon\right) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2 + \frac{2}{3}M\varepsilon}\right).$$

In our case, we have already shown that $\mathbb{E}[Y_i] = 0$, $|Y_i| \leq 2$, and $\text{Var}(Y_i) \leq 2$. Thus we may take

$$M = 2, \quad \sigma^2 = 2.$$

Since

$$G_{pq} - \delta_{pq} = \frac{1}{n} \sum_{i=1}^n Y_i,$$

Bernstein's inequality for the sample mean yields

$$\mathbb{P}(|G_{pq} - \delta_{pq}| \geq \varepsilon) \leq 2 \exp\left(-\frac{n\varepsilon^2}{4 + \frac{4}{3}\varepsilon}\right).$$

Since $G - I_d$ is symmetric, it suffices to union bound over the $d(d+1)/2$ entries in the upper triangle. Therefore

$$\mathbb{P}(\|G - I_d\|_{\max} \geq \varepsilon) \leq d(d+1) \exp\left(-\frac{n\varepsilon^2}{4 + \frac{4}{3}\varepsilon}\right),$$

which proves (4.13). □

C.4 Concentration of H

Proof of (4.14). Fix p, q , and write $H_{pq} = F(\varphi_1, \dots, \varphi_n)$, where

$$F(\varphi_1, \dots, \varphi_n) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \phi_p(\varphi_i) \Theta(\varphi_i - \varphi_j) \phi_q(\varphi_j).$$

We apply McDiarmid's inequality (Vershynin, 2018b) in the following form: if $X_1, \dots, X_n$ are independent and $F(X_1, \dots, X_n)$ satisfies the bounded-differences condition

$$|F(x_1, \dots, x_s, \dots, x_n) - F(x_1, \dots, x'_s, \dots, x_n)| \leq c_s$$

for each s , then for every $\varepsilon > 0$,

$$\mathbb{P}(|F - \mathbb{E}F| \geq \varepsilon) \leq 2 \exp\left(-\frac{2\varepsilon^2}{\sum_{s=1}^n c_s^2}\right).$$

It therefore remains to estimate the effect of changing a single sample point, say $\varphi_s \mapsto \varphi'_s$, while keeping all others fixed. Only those summands in the double sum with $i = s$ or $j = s$ can change. There are at most $2n$ such summands.

By (C.1) and (C.2), each summand satisfies

$$|\phi_p(\cdot) \Theta(\cdot) \phi_q(\cdot)| \leq \sqrt{2} \kappa \sqrt{2} = 2\kappa.$$

Hence, when φ_s is replaced by φ'_s , a single summand can change by at most 4κ .

Because at most $2n$ summands are affected, the total change in the double sum is bounded by

$$(2n) \cdot (4\kappa) = 8\kappa n.$$

Finally, since F contains the prefactor $1/n^2$, the change in F itself is bounded by

$$\frac{8\kappa n}{n^2} = \frac{8\kappa}{n}.$$

Thus we may take

$$c_s = \frac{8\kappa}{n} \quad \text{for all } s = 1, \dots, n.$$

Therefore

$$\sum_{s=1}^n c_s^2 = n \left(\frac{8\kappa}{n}\right)^2 = \frac{64\kappa^2}{n}.$$

McDiarmid's inequality yields

$$\mathbb{P}(|H_{pq} - \mathbb{E}[H_{pq}]| \geq \varepsilon) \leq 2 \exp\left(-\frac{2\varepsilon^2}{64\kappa^2/n}\right) = 2 \exp\left(-\frac{n\varepsilon^2}{32\kappa^2}\right).$$

A union bound over all d^2 entries gives

$$\mathbb{P}\left(\|H - \Lambda^{(n)}\|_{\max} \geq \varepsilon\right) \leq 2d^2 \exp\left(-\frac{n\varepsilon^2}{32\kappa^2}\right),$$

as claimed. □

Remark C.1. The entry H_{pq} is a bounded second-order V-statistic, so Bernstein-type refinements are in principle available through concentration inequalities for bounded U/V-statistics. Since the McDiarmid bound already captures the correct $n^{-1/2}$ concentration scale and is sufficient for the results in this thesis, we do not pursue this refinement here.

C.5 Control of the difference $A\Phi - \Phi\Lambda^{(n)}$

Proof of (4.15). Fix i, p . By definition,

$$(A\Phi)_{i,p} = \frac{1}{n} \sum_{j=1}^n \Theta(\varphi_i - \varphi_j) \phi_p(\varphi_j).$$

Split off the diagonal term $j = i$:

$$(A\Phi)_{i,p} = \frac{\Theta(0)}{n} \phi_p(\varphi_i) + \frac{1}{n} \sum_{j \neq i} \Theta(\varphi_i - \varphi_j) \phi_p(\varphi_j).$$

Condition on φ_i . For $j \neq i$, define

$$X_j := \Theta(\varphi_i - \varphi_j) \phi_p(\varphi_j).$$

Then $\{X_j\}_{j \neq i}$ are conditionally independent and identically distributed.

Their conditional mean is

$$\mathbb{E}[X_j \mid \varphi_i] = \int_0^{2\pi} \Theta(\varphi_i - \psi) \phi_p(\psi) d\mu(\psi) = (T\phi_p)(\varphi_i) = \lambda_p \phi_p(\varphi_i).$$

Therefore

$$\mathbb{E}[(A\Phi)_{i,p} \mid \varphi_i] = \frac{\Theta(0)}{n} \phi_p(\varphi_i) + \frac{n-1}{n} \lambda_p \phi_p(\varphi_i) = \lambda_p^{(n)} \phi_p(\varphi_i).$$

Now set

$$Y_j := X_j - \mathbb{E}[X_j \mid \varphi_i].$$

Conditional on φ_i , the variables $\{Y_j\}_{j \neq i}$ are independent, centered, and identically distributed.

We apply Bernstein's inequality (Vershynin, 2018a) in the following form: if $Y_1, \dots, Y_m$ are independent, centered random variables such that

$$|Y_j| \leq M, \quad \text{Var}(Y_j) \leq \sigma^2$$

for all j , then for every $\eta > 0$,

$$\mathbb{P}\left(\left|\frac{1}{m} \sum_{j=1}^m Y_j\right| \geq \eta\right) \leq 2 \exp\left(-\frac{m\eta^2}{2\sigma^2 + \frac{2}{3}M\eta}\right).$$

We now verify the assumptions conditionally on φ_i .

First, by (C.1) and (C.2),

$$|X_j| \leq \kappa\sqrt{2} \leq 2\kappa.$$

Also,

$$|\mathbb{E}[X_j \mid \varphi_i]| = \left| \int_0^{2\pi} \Theta(\varphi_i - \psi) \phi_p(\psi) d\mu(\psi) \right| \leq \kappa \int_0^{2\pi} |\phi_p(\psi)| d\mu(\psi) \leq \kappa \|\phi_p\|_{L^2(\mu)} = \kappa.$$

Hence

$$|Y_j| = |X_j - \mathbb{E}[X_j \mid \varphi_i]| \leq 2\kappa + \kappa = 3\kappa.$$

Next,

$$\text{Var}(Y_j \mid \varphi_i) = \text{Var}(X_j \mid \varphi_i) \leq \mathbb{E}[X_j^2 \mid \varphi_i].$$

Using again (C.1) and (C.2),

$$\mathbb{E}[X_j^2 \mid \varphi_i] = \int_0^{2\pi} \Theta(\varphi_i - \psi)^2 \phi_p(\psi)^2 d\mu(\psi) \leq \kappa^2 \int_0^{2\pi} \phi_p(\psi)^2 d\mu(\psi) = \kappa^2.$$

Thus, conditionally on φ_i , Bernstein's inequality applies with

$$M = 3\kappa, \quad \sigma^2 = \kappa^2.$$

Let $m := n - 1$. Applying Bernstein to the conditional sample mean

$$\frac{1}{m} \sum_{j \neq i} Y_j = \frac{1}{m} \sum_{j \neq i} X_j - \lambda_p \phi_p(\varphi_i),$$

we obtain

$$\mathbb{P} \left(\left| \frac{1}{m} \sum_{j \neq i} X_j - \lambda_p \phi_p(\varphi_i) \right| \geq \eta \mid \varphi_i \right) \leq 2 \exp \left(-\frac{m\eta^2}{2\kappa^2 + 2\kappa\eta} \right).$$

Now

$$(A\Phi)_{i,p} - \lambda_p^{(n)} \phi_p(\varphi_i) = \frac{m}{n} \left(\frac{1}{m} \sum_{j \neq i} X_j - \lambda_p \phi_p(\varphi_i) \right).$$

Thus the event

$$\left| (A\Phi)_{i,p} - \lambda_p^{(n)} \phi_p(\varphi_i) \right| \geq \varepsilon$$

implies

$$\left| \frac{1}{m} \sum_{j \neq i} X_j - \lambda_p \phi_p(\varphi_i) \right| \geq \frac{n}{m} \varepsilon.$$

Hence

$$\mathbb{P} \left(\left| (A\Phi)_{i,p} - \lambda_p^{(n)} \phi_p(\varphi_i) \right| \geq \varepsilon \mid \varphi_i \right) \leq 2 \exp \left(-\frac{m \left(\frac{n}{m} \varepsilon \right)^2}{2\kappa^2 + 2\kappa \left(\frac{n}{m} \varepsilon \right)} \right).$$

Removing the conditioning and simplifying gives

$$\mathbb{P} \left(\left| (A\Phi)_{i,p} - \lambda_p^{(n)} \phi_p(\varphi_i) \right| \geq \varepsilon \right) \leq 2 \exp \left(-\frac{n^2 \varepsilon^2}{2\kappa^2(n-1) + 2\kappa n \varepsilon} \right).$$

Since $n - 1 \leq n$, we obtain

$$\mathbb{P}\left(\left|(A\Phi)_{i,p} - \lambda_p^{(n)} \phi_p(\varphi_i)\right| \geq \varepsilon\right) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\kappa^2 + 2\kappa\varepsilon}\right).$$

Finally, taking a union bound over all nd pairs (i, p) yields

$$\mathbb{P}\left(\|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \geq \varepsilon\right) \leq 2nd \exp\left(-\frac{n\varepsilon^2}{2\kappa^2 + 2\kappa\varepsilon}\right),$$

which proves (4.15). $\square$

C.6 Proof of the simultaneous high-probability bound

Proof of Corollary 4.6. Set

$$u_{n,d,\delta}^{(G)} = \log\left(\frac{3d(d+1)}{\delta}\right), \quad u_{n,d,\delta}^{(H)} = \log\left(\frac{6d^2}{\delta}\right), \quad u_{n,d,\delta}^{(A)} = \log\left(\frac{6nd}{\delta}\right).$$

Define

$$\alpha_{n,d,\delta} = 2\sqrt{\frac{u_{n,d,\delta}^{(G)}}{n}} + \frac{2}{3} \frac{u_{n,d,\delta}^{(G)}}{n},$$

$$\beta_{n,d,\delta} = \kappa\sqrt{\frac{32}{n} u_{n,d,\delta}^{(H)}}, \quad \gamma_{n,d,\delta} = \kappa\sqrt{\frac{2}{n} u_{n,d,\delta}^{(A)}} + \kappa \frac{u_{n,d,\delta}^{(A)}}{n}.$$

We first recall the standard Bernstein inversion: if

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n Y_i\right| \geq t\right) \leq 2 \exp\left(-\frac{nt^2}{2\sigma^2 + \frac{2}{3}Mt}\right),$$

then choosing

$$t = \sqrt{\frac{2\sigma^2 u}{n}} + \frac{Mu}{3n}$$

ensures

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n Y_i\right| \geq t\right) \leq 2e^{-u}.$$

For the Gram matrix bound (4.13), the Bernstein proof used

$$\sigma^2 = 2, \quad M = 2.$$

Thus, with

$$u = u_{n,d,\delta}^{(G)},$$

the choice

$$\alpha_{n,d,\delta} = \sqrt{\frac{2 \cdot 2u}{n}} + \frac{2u}{3n} = 2\sqrt{\frac{u}{n}} + \frac{2u}{3n}$$

yields

$$\mathbb{P}(\|G - I_d\|_{\max} \geq \alpha_{n,d,\delta}) \leq d(d+1)e^{-u_{n,d,\delta}^{(G)}} = \frac{\delta}{3}.$$

For the compressed operator, the McDiarmid estimate (4.14) gives

$$\mathbb{P}(\|H - \Lambda^{(n)}\|_{\max} \geq \varepsilon) \leq 2d^2 \exp\left(-\frac{n\varepsilon^2}{32\kappa^2}\right).$$

Substituting $\varepsilon = \beta_{n,d,\delta}$ gives

$$\frac{n\beta_{n,d,\delta}^2}{32\kappa^2} = u_{n,d,\delta}^{(H)},$$

and therefore

$$\mathbb{P}(\|H - \Lambda^{(n)}\|_{\max} \geq \beta_{n,d,\delta}) \leq 2d^2 e^{-u_{n,d,\delta}^{(H)}} = \frac{\delta}{3}.$$

For the approximate eigen-action bound (4.15), the Bernstein proof used

$$\sigma^2 = \kappa^2, \quad M = 3\kappa.$$

Hence, with

$$u = u_{n,d,\delta}^{(A)},$$

the standard Bernstein inversion gives the threshold

$$\sqrt{\frac{2\kappa^2 u}{n}} + \frac{3\kappa u}{3n} = \kappa \sqrt{\frac{2u}{n}} + \kappa \frac{u}{n},$$

which is precisely $\gamma_{n,d,\delta}$. Therefore

$$\mathbb{P}(\|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \geq \gamma_{n,d,\delta}) \leq 2nd e^{-u_{n,d,\delta}^{(A)}} = \frac{\delta}{3}.$$

A final union bound gives

$$\mathbb{P}(\|G - I_d\|_{\max} \leq \alpha_{n,d,\delta}, \|H - \Lambda^{(n)}\|_{\max} \leq \beta_{n,d,\delta}, \|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \leq \gamma_{n,d,\delta}) \geq 1 - \delta,$$

which is exactly (4.22). $\square$

C.7 Proof of the corollary on the restricted sampled operator

Appendix D

Proofs of the finite-width frozen-kernel estimates

This appendix proves the finite-width frozen-kernel results stated in Section 4.4. We use the notation introduced in Chapter 3. In particular, $T_0^{(m)}$ denotes the frozen finite-width tangent operator at initialization, T_∞ the infinite-width limiting operator, $\mathcal{H}_K$ the truncated Fourier subspace, P_K the orthogonal projection onto $\mathcal{H}_K$, and

$$B_m^{(K)} = P_K T_0^{(m)} P_K, \quad \Lambda_K = P_K T_\infty P_K.$$

Since the input space is the continuum S^1 , there is no sampling randomness in this appendix. The only randomness comes from the finite number of initialized neurons.

D.1 One-neuron tangent feature and kernel

Fix one neuron r , and write

$$u_r(\theta) := w_r^\top x(\theta) + b_r.$$

At initialization, $b_r = 0$, so $u_r(\theta) = w_r^\top x(\theta)$.

We compute the tangent feature of the r -th neuron contribution

$$a_r \sigma(u_r(\theta))$$

with respect to the parameter block (a_r, w_r, b_r) . First,

$$\partial_{a_r}(a_r \sigma(u_r(\theta))) = \sigma(u_r(\theta)).$$

Next, by the chain rule,

$$\nabla_{w_r}(a_r \sigma(u_r(\theta))) = a_r \sigma'(u_r(\theta)) \nabla_{w_r} u_r(\theta).$$

Since

$$u_r(\theta) = w_r^\top x(\theta) + b_r,$$

we have

$$\nabla_{w_r} u_r(\theta) = x(\theta).$$

For ReLU,

$$\sigma'(z) = \mathbf{1}_{\{z>0\}}$$

away from $z = 0$, which is sufficient almost surely under the Gaussian initialization. Hence

$$\nabla_{w_r} (a_r \sigma(u_r(\theta))) = a_r \mathbf{1}_{\{u_r(\theta)>0\}} x(\theta).$$

Similarly,

$$\partial_{b_r} (a_r \sigma(u_r(\theta))) = a_r \sigma'(u_r(\theta)) \partial_{b_r} u_r(\theta) = a_r \mathbf{1}_{\{u_r(\theta)>0\}}.$$

Thus the one-neuron tangent feature is

$$\Psi_r(\theta) = \begin{pmatrix} \sigma(u_r(\theta)) \\ a_r \mathbf{1}_{\{u_r(\theta)>0\}} \cos \theta \\ a_r \mathbf{1}_{\{u_r(\theta)>0\}} \sin \theta \\ a_r \mathbf{1}_{\{u_r(\theta)>0\}} \end{pmatrix} \in \mathbb{R}^4. \quad (\text{D.1})$$

The corresponding one-neuron kernel is

$$\Theta^{(r)}(\theta, \eta) = \Psi_r(\theta)^\top \Psi_r(\eta). \quad (\text{D.2})$$

Substituting (D.1) gives

$$\begin{aligned} \Theta^{(r)}(\theta, \eta) &= \sigma(u_r(\theta)) \sigma(u_r(\eta)) \\ &\quad + a_r^2 \mathbf{1}_{\{u_r(\theta)>0\}} \mathbf{1}_{\{u_r(\eta)>0\}} (\cos \theta \cos \eta + \sin \theta \sin \eta + 1). \end{aligned} \quad (\text{D.3})$$

Using

$$\cos \theta \cos \eta + \sin \theta \sin \eta = \cos(\theta - \eta),$$

we obtain

$$\Theta^{(r)}(\theta, \eta) = \sigma(u_r(\theta)) \sigma(u_r(\eta)) + a_r^2 \mathbf{1}_{\{u_r(\theta)>0\}} \mathbf{1}_{\{u_r(\eta)>0\}} (\cos(\theta - \eta) + 1). \quad (\text{D.4})$$

Let $T^{(r)}$ denote the associated one-neuron integral operator,

$$(T^{(r)} g)(\theta) := \int_0^{2\pi} \Theta^{(r)}(\theta, \eta) g(\eta) d\mu(\eta). \quad (\text{D.5})$$

Then, by (3.39),

$$T_0^{(m)} = \frac{1}{m} \sum_{r=1}^m T^{(r)}.$$

D.2 Projected one-neuron matrices

Using the ordered basis

$$(\phi_0, \phi_{1,c}, \phi_{1,s}, \dots, \phi_{K,c}, \phi_{K,s})$$

of $\mathcal{H}_K$, the entries of $B_m^{(K)}$ are

$$(B_m^{(K)})_{pq} = \langle \phi_p, T_0^{(m)} \phi_q \rangle_{L^2(\mu)}. \quad (\text{D.6})$$

Since $T_0^{(m)} = \frac{1}{m} \sum_{r=1}^m T^{(r)}$,

$$(B_m^{(K)})_{pq} = \frac{1}{m} \sum_{r=1}^m \langle \phi_p, T^{(r)} \phi_q \rangle.$$

Define

$$\xi_{r,pq} := \langle \phi_p, T^{(r)} \phi_q \rangle. \quad (\text{D.7})$$

Then

$$(B_m^{(K)})_{pq} = \frac{1}{m} \sum_{r=1}^m \xi_{r,pq}. \quad (\text{D.8})$$

Equivalently, if Z_r denotes the $d \times d$ random matrix defined by

$$(Z_r)_{pq} := \xi_{r,pq},$$

then

$$B_m^{(K)} = \frac{1}{m} \sum_{r=1}^m Z_r. \quad (\text{D.9})$$

Since the neurons are initialized independently and identically, the matrices $Z_1, \dots, Z_m$ are i.i.d.

D.3 Expectation of the projected operator

Proof of (4.25). From (D.9),

$$\mathbb{E}[B_m^{(K)}] = \frac{1}{m} \sum_{r=1}^m \mathbb{E}[Z_r] = \mathbb{E}[Z_1].$$

The (p, q) -entry is

$$\mathbb{E}[(Z_1)_{pq}] = \mathbb{E}[\xi_{1,pq}] = \mathbb{E}[\langle \phi_p, T^{(1)} \phi_q \rangle].$$

By linearity of expectation,

$$\mathbb{E}[\xi_{1,pq}] = \langle \phi_p, \mathbb{E}[T^{(1)}] \phi_q \rangle = \langle \phi_p, T_\infty \phi_q \rangle.$$

Since T_∞ is diagonal in the Fourier basis,

$$\langle \phi_p, T_\infty \phi_q \rangle = \lambda_q \delta_{pq}.$$

Therefore

$$\mathbb{E}[B_m^{(K)}] = \Lambda_K.$$

□

D.4 Sub-exponential bound for one projected entry

We next control $\xi_{r,pq}$. By definition,

$$\xi_{r,pq} = \int_0^{2\pi} \int_0^{2\pi} \phi_p(\theta) \Theta^{(r)}(\theta, \eta) \phi_q(\eta) d\mu(\eta) d\mu(\theta). \quad (\text{D.10})$$

Taking absolute values gives

$$|\xi_{r,pq}| \leq \int_0^{2\pi} \int_0^{2\pi} |\phi_p(\theta)| |\Theta^{(r)}(\theta, \eta)| |\phi_q(\eta)| d\mu(\eta) d\mu(\theta).$$

Since every retained Fourier basis function satisfies

$$\|\phi_p\|_\infty \leq \sqrt{2},$$

we obtain

$$|\xi_{r,pq}| \leq 2 \int_0^{2\pi} \int_0^{2\pi} |\Theta^{(r)}(\theta, \eta)| d\mu(\eta) d\mu(\theta).$$

We now bound the kernel. Since

$$u_r(\theta) = w_r^\top x(\theta), \quad \|x(\theta)\| = 1,$$

Cauchy–Schwarz gives

$$|u_r(\theta)| \leq \|w_r\|.$$

Therefore

$$0 \leq \sigma(u_r(\theta)) \leq |u_r(\theta)| \leq \|w_r\|,$$

and hence

$$0 \leq \sigma(u_r(\theta)) \sigma(u_r(\eta)) \leq \|w_r\|^2.$$

For the derivative and bias term, we use

$$\mathbf{1}_{\{u_r(\theta) > 0\}} \mathbf{1}_{\{u_r(\eta) > 0\}} \leq 1$$

and

$$0 \leq \cos(\theta - \eta) + 1 \leq 2.$$

Thus

$$a_r^2 \mathbf{1}_{\{u_r(\theta) > 0\}} \mathbf{1}_{\{u_r(\eta) > 0\}} (\cos(\theta - \eta) + 1) \leq 2a_r^2.$$

Using (D.4), we conclude that

$$|\Theta^{(r)}(\theta, \eta)| \leq \|w_r\|^2 + 2a_r^2.$$

Since μ is a probability measure,

$$\int \int (\|w_r\|^2 + 2a_r^2) d\mu(\eta) d\mu(\theta) = \|w_r\|^2 + 2a_r^2.$$

Therefore

$$|\xi_{r,pq}| \leq 2\|w_r\|^2 + 4a_r^2. \quad (\text{D.11})$$

Define

$$Y_r := 2\|w_r\|^2 + 4a_r^2.$$

Then

$$|\xi_{r,pq}| \leq Y_r \quad \text{for all } p, q.$$

We now show that Y_r is sub-exponential with an explicit parameter. Since $w_r \sim \mathcal{N}(0, I_2)$, its squared Euclidean norm is the sum of two independent standard Gaussian squares, and therefore w_r follows a chi-squared distribution with two degrees of freedom $\|w_r\|^2 \sim \chi_2^2$.

Similarly, since $a_r \sim \mathcal{N}(0, 1)$, its square follows a chi-squared distribution with one degree of freedom $a_r^2 \sim \chi_1^2$ and these variables are independent, for $t < 1/8$,

$$\mathbb{E}[e^{tY_r}] = \mathbb{E}[e^{2t\|w_r\|^2}] \mathbb{E}[e^{4ta_r^2}].$$

If $X \sim \chi_v^2$, then

$$\mathbb{E}[e^{sX}] = (1 - 2s)^{-v/2}, \quad s < 1/2.$$

Hence

$$\mathbb{E}[e^{2t\|w_r\|^2}] = (1 - 4t)^{-1}, \quad \mathbb{E}[e^{4ta_r^2}] = (1 - 8t)^{-1/2}.$$

Thus

$$\mathbb{E}[e^{tY_r}] = (1 - 4t)^{-1} (1 - 8t)^{-1/2}, \quad t < 1/8. \quad (\text{D.12})$$

Choosing $t = 1/16$, we obtain

$$\mathbb{E}[e^{Y_r/16}] = \left(\frac{3}{4}\right)^{-1} \left(\frac{1}{2}\right)^{-1/2} = \frac{4}{3}\sqrt{2} < 2.$$

Therefore, by the standard definition of the ψ_1 -norm,

$$\|Y_r\|_{\psi_1} \leq 16. \quad (\text{D.13})$$

Since $|\xi_{r,pq}| \leq Y_r$,

$$\|\xi_{r,pq}\|_{\psi_1} \leq 16.$$

Finally,

$$\|\xi_{r,pq} - \mathbb{E}[\xi_{r,pq}]\|_{\psi_1} \leq \|\xi_{r,pq}\|_{\psi_1} + |\mathbb{E}[\xi_{r,pq}]|.$$

Using $|\mathbb{E}[\xi_{r,pq}]| \leq \mathbb{E}|\xi_{r,pq}|$ and $\mathbb{E}|X| \leq \|X\|_{\psi_1}$, we obtain

$$\|\xi_{r,pq} - \mathbb{E}[\xi_{r,pq}]\|_{\psi_1} \leq 32. \quad (\text{D.14})$$

D.5 Entrywise and blockwise concentration

Proof of (4.26). Fix p, q , and define

$$X_{r,pq} := \xi_{r,pq} - \mathbb{E}[\xi_{r,pq}].$$

Then $X_{1,pq}, \dots, X_{m,pq}$ are i.i.d., centered, and by (D.14),

$$\|X_{r,pq}\|_{\Psi_1} \leq 32.$$

Moreover,

$$(B_m^{(K)})_{pq} - (\Lambda_K)_{pq} = \frac{1}{m} \sum_{r=1}^m X_{r,pq}.$$

We apply the standard Bernstein inequality for centered i.i.d. sub-exponential random variables: if $X_1, \dots, X_m$ satisfy $\|X_i\|_{\Psi_1} \leq L$, then there exists a universal constant $c > 0$ such that

$$\mathbb{P}\left(\left|\frac{1}{m} \sum_{r=1}^m X_r\right| \geq \varepsilon\right) \leq 2 \exp\left(-cm \min\left(\frac{\varepsilon^2}{L^2}, \frac{\varepsilon}{L}\right)\right).$$

Applying this with $L = 32$, we obtain

$$\mathbb{P}\left(\left|(B_m^{(K)})_{pq} - (\Lambda_K)_{pq}\right| \geq \varepsilon\right) \leq 2 \exp\left(-cm \min\left(\frac{\varepsilon^2}{32^2}, \frac{\varepsilon}{32}\right)\right). \quad (\text{D.15})$$

There are $d^2 = (2K+1)^2$ entries. By the union bound,

$$\mathbb{P}\left(\|B_m^{(K)} - \Lambda_K\|_{\max} \geq \varepsilon\right) \leq \sum_{p,q=1}^d \mathbb{P}\left(\left|(B_m^{(K)})_{pq} - (\Lambda_K)_{pq}\right| \geq \varepsilon\right).$$

Using (D.15),

$$\mathbb{P}\left(\|B_m^{(K)} - \Lambda_K\|_{\max} \geq \varepsilon\right) \leq 2d^2 \exp\left(-cm \min\left(\frac{\varepsilon^2}{32^2}, \frac{\varepsilon}{32}\right)\right).$$

Since $d = 2K+1$, this proves (4.26). $\square$